Ministry Approved
2012

National Textbook Award

Compulsory Education Textbook

Mathematics

SHUXUE

Grade 1 Volume 2

北京师范大学出版社

Beijing Normal University Press
BEIJING NORMAL UNIVERSITY PRESS

Dear children,

After a semester of maths, do you think maths is fun?

Wise Old Man

I like numbers. They are everywhere in life.

Taoqi

56 56 | 565
55 3 | 55 34 | 55 3

I drew 4 big watermelons.

How many pretty fish?

I like shapes. I used shapes to draw a big maths castle.

Xiaoxiao

Woof! There's maths in here!

Clever Dog

Children, do you think maths is fun? Share your ideas with everyone.
We believe the lessons ahead will bring you even more joy.

Your friends, the editors

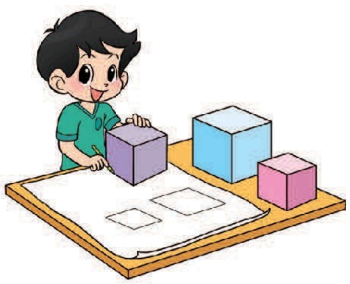
浙江大学出版社

Contents

Unit 1 Addition and Subtraction (1) ····· 2

Unit 2 Observing Objects ····· 18

Unit 3 Numbers in Life ····· 22



Unit 4 Interesting Shapes ····· 36

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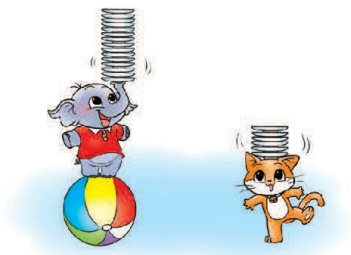
Review ····· 44



Unit 5 Addition and Subtraction (2) ····· 49

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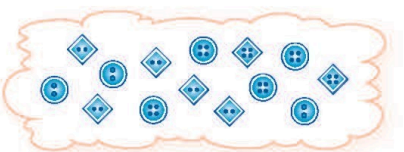
Maths Is Fun ····· 64



Unit 6 Addition and Subtraction (3) ····· 68

Final Review ····· 84

Unit 1 Addition and Subtraction (1)

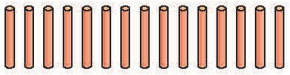
Buying Pencils

How many pencils are left?

□ ○ □ = □

We can use sticks.

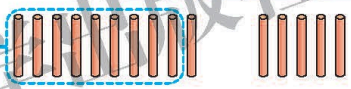
Take away one at a time, leaving...



Split 15 into 10 and 5

$10-9=1$

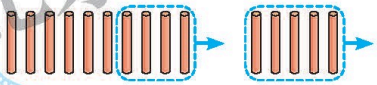
$1+5=6$



Split 9 into 5 and 4

$15-5=10$

$10-4=6$



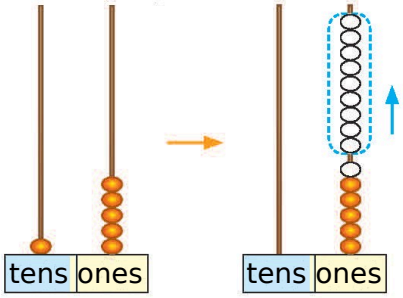
9 and 6 more makes 15.

$9+6=15$

$15-9=6$

Answer: ___ pencils are left.

Try it, then explain.



Is there another way?

Calculate.

$12-9=\square$ $17-9=\square$

Calculate in your favourite way.

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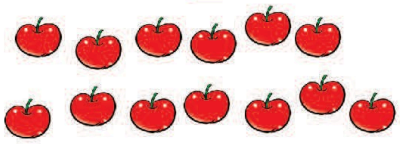
Practice

1. Use sticks or a counting frame, then calculate.

$11-9=$ $18-9=$

2. Circle and calculate.

$13-9=\square$



$16-9=\square$

3.

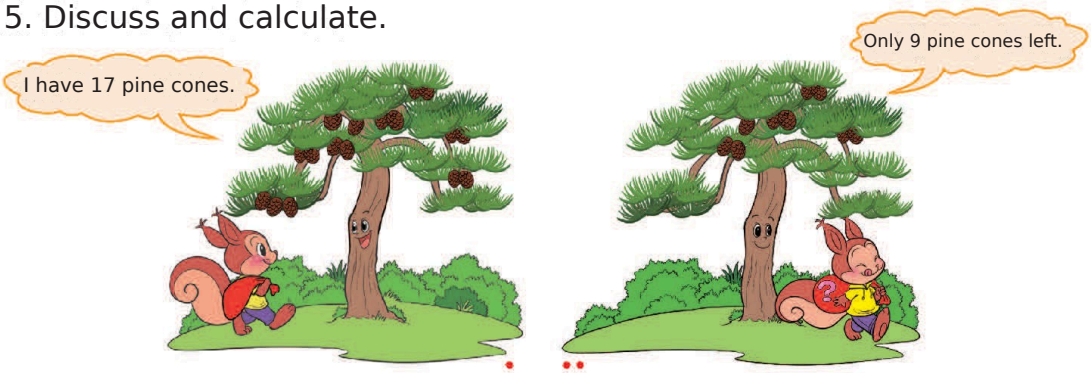
$9+5=$

$14-9=$

4.

$10-9=$	$11-9=$	$16-9=$	$13-9=$
$18-9=$	$19-9=$	$12-9=$	$15-9=$

5. Discuss and calculate.



$\square-\square=\square$

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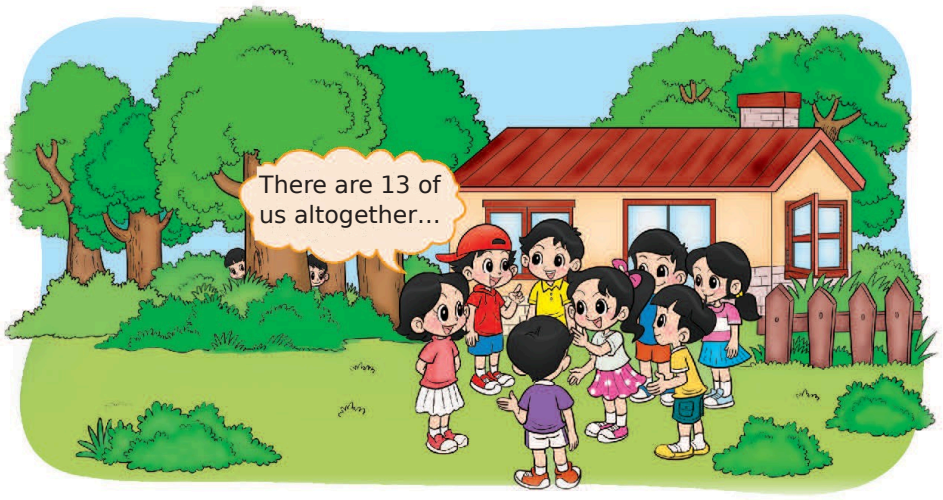
Hide and Seek



How many children are hiding?

□ ○ □ = □

10 − 8 = 2
2 + 3 = 5

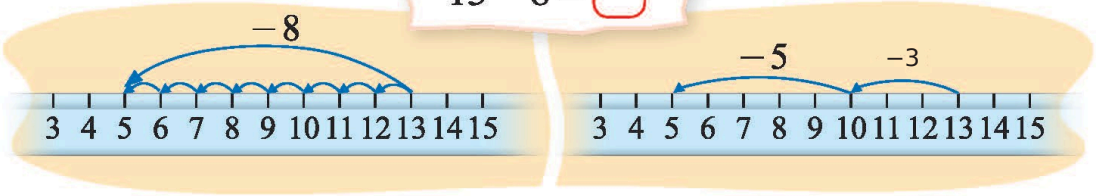
13 − 3 = 10
10 − 5 = 5

8 + 5 = 13
13 − 8 = 5

Answer: ____ children are hiding.

Look and talk about it.

13 − 8 = □



Calculate.

12 − 8 = □　　17 − 8 = □

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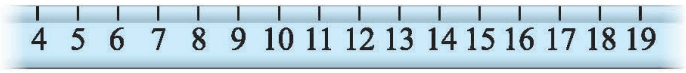
Practice

1. Use sticks or a counting frame, then calculate.

$11-8=$ $14-8=$

2. Draw and fill in.

$16-8=\square$



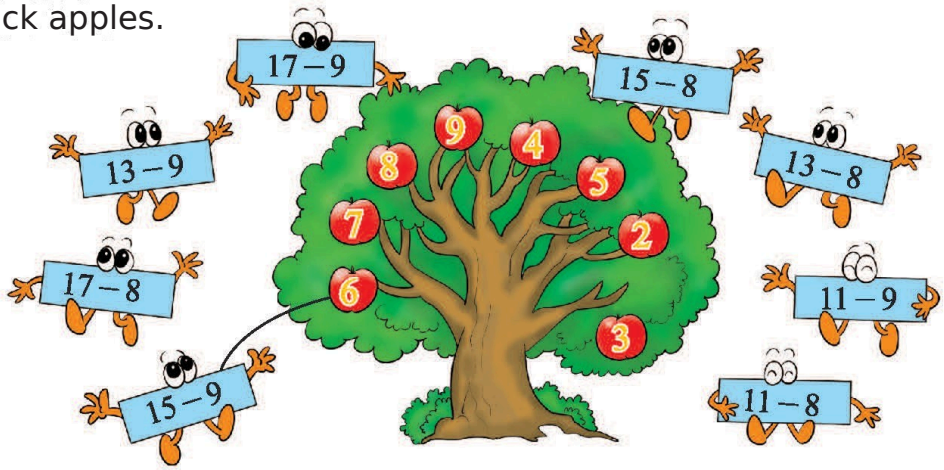
3.

$\square-\square=\square$ $\square-\square=\square$

4.

$13-9=$	$14-8=$	$17-7=$
$14-9=$	$15-8=$	$17-8=$
$15-9=$	$16-8=$	$17-9=$

5. Pick apples.



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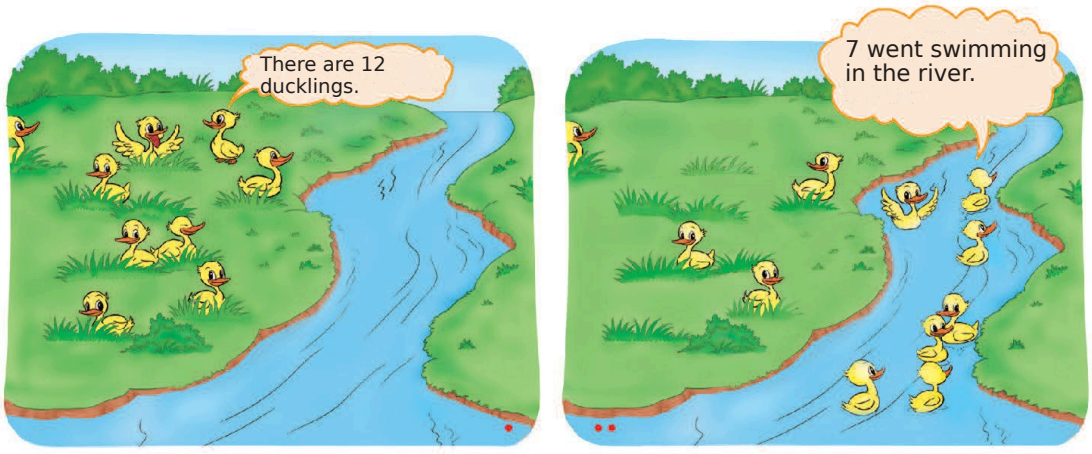
Happy Ducklings



How many are still on the grass?

□ ○ □ = □

$12-7=5$

10 2

$12-7=5$

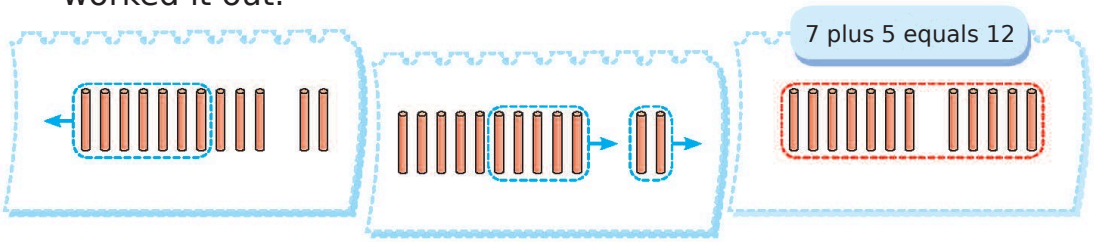
2 5

$7+5=12$

$12-7=5$

Answer: ___ ducklings are still on the grass.

Use sticks to show it, and explain how you worked it out.



Work it out with sticks.

$11-6=\square$ $14-6=\square$

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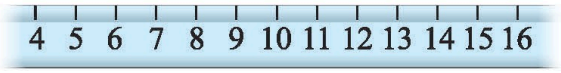
Practice

1. Calculate. Use sticks or a counting frame to show how you worked it out.

$14-7=$ $12-6=$

2. Draw and fill in.

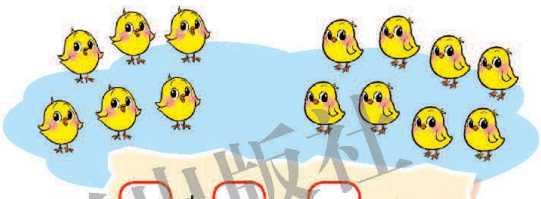
$13-6=\square$



3.

$7+4=\square$

$11-7=\square$



$\square+\square=\square$

$\square-\square=\square$

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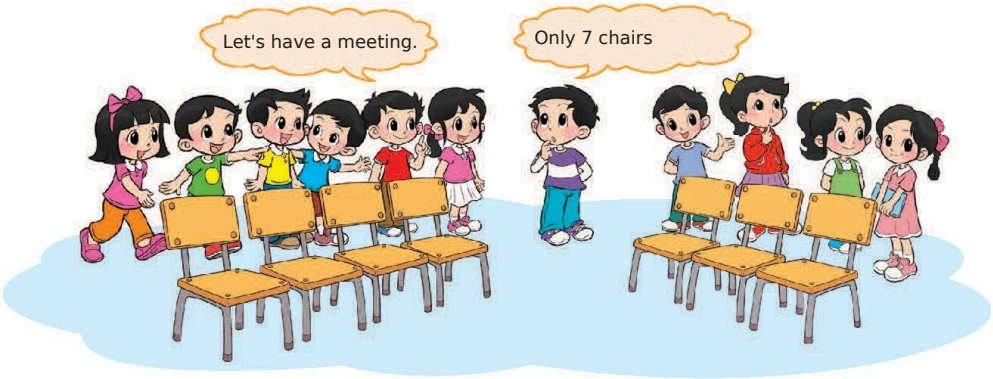
4.

$15-6=$	$13-7=$	$16-7=$	$15-7=$
$12-7=$	$11-6=$	$14-6=$	$13-6=$

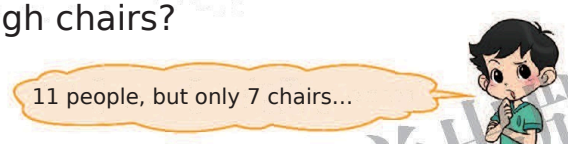
5. Little Bear shoots at goal.

$\square\bigcirc\square=\square$

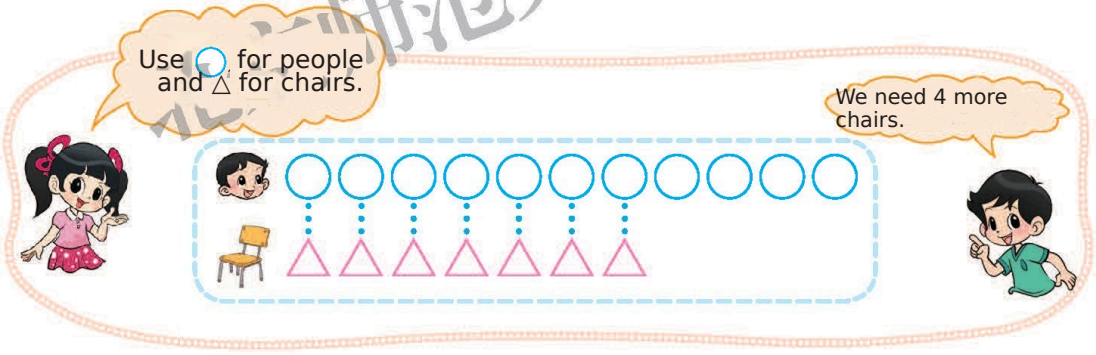
Meeting



If each person sits on 1 chair, are there enough chairs?

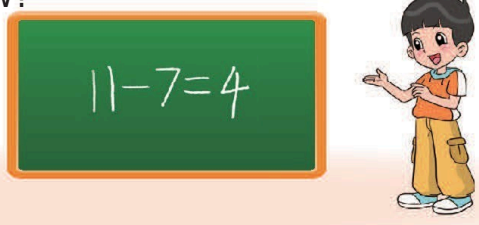


How many more chairs are needed?



Answer: ___ more chairs are needed.

Can you write a number sentence to solve it? Do you agree with the one below?



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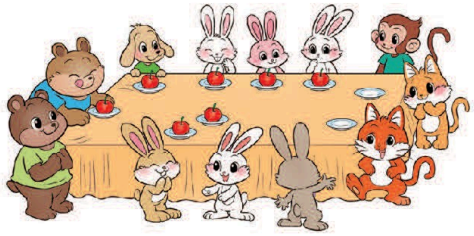
Practice

1. Each child gets 1 shovel. How many more shovels are needed?

2.



(1) If each animal gets 1 apple, how many more apples are needed?

(2) Ask another maths question and solve it.

3.

$12-8=$	$14-9=$	$15-9=$	$15-6=$
$17-8=$	$13-7=$	$11-8=$	$14-6=$

4. ☐ What goes in each box?

$8+\square=14$	$6+\square=13$	$\square+9=11$
$14-8=\square$	$13-6=\square$	$11-9=\square$

5. Fill in.

$11-6=5$

11	12	13	14	15	16	17
5						

−6

$17-7=10$

11	12	13	14	15	16	17
						10

−7

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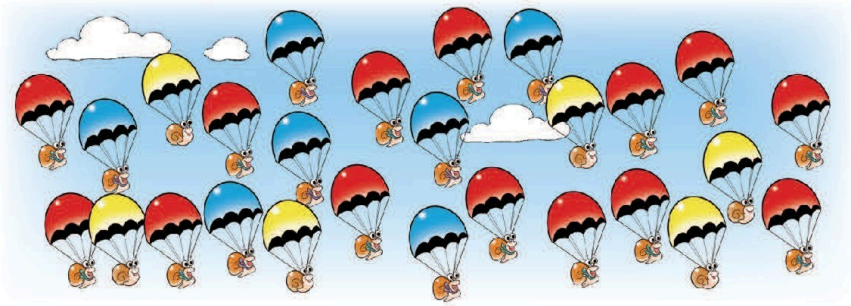
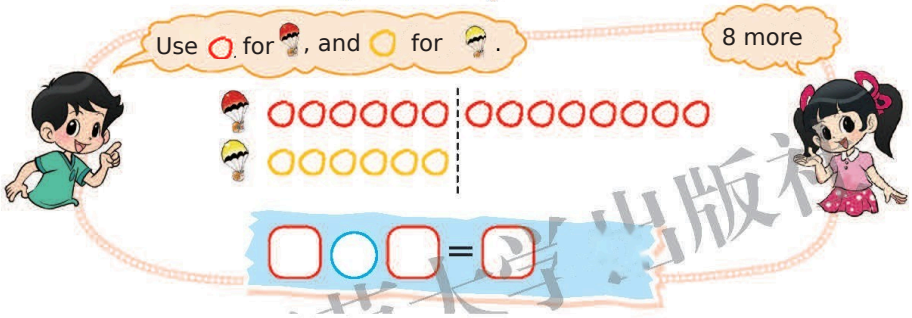
Parachutes

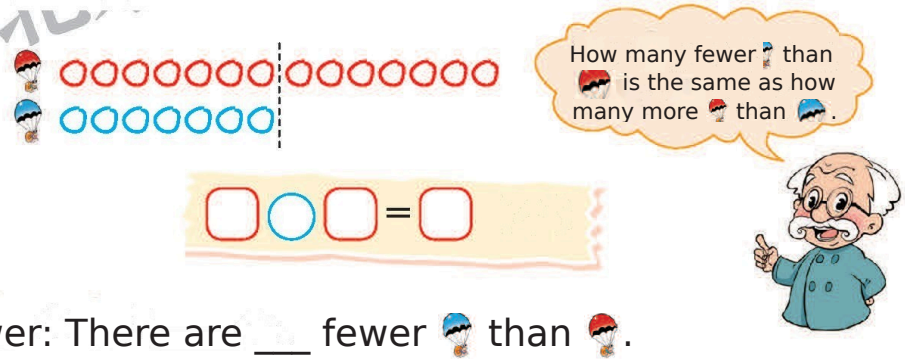


How many more 🪂 than 🪂?



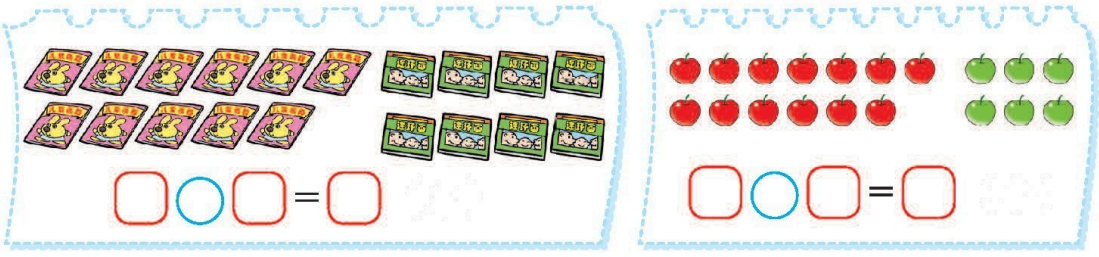
Answer: There are ___ more red parachutes than yellow ones.

How many fewer blue parachutes than red ones?



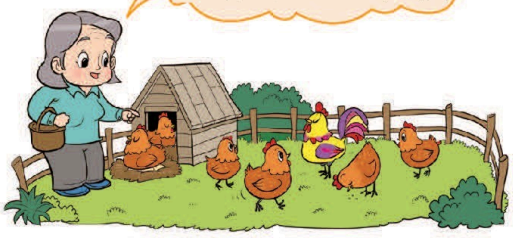
Answer: There are ___ fewer 🪂 than 🪂.

Compare and calculate.

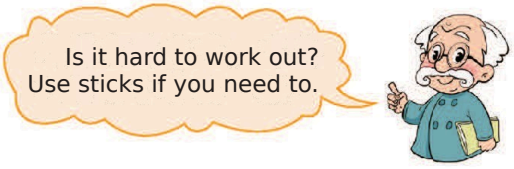


Try It

How many fewer eggs were laid today than yesterday?



□ ○ □ = □



Answer: ___ fewer eggs were laid today than yesterday.

Calculate. What do you notice?

$11-2=\square$ $12-3=\square$ $13-4=\square$

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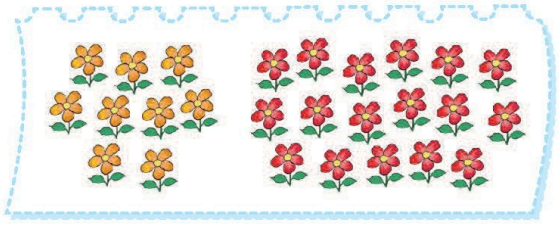
Practice

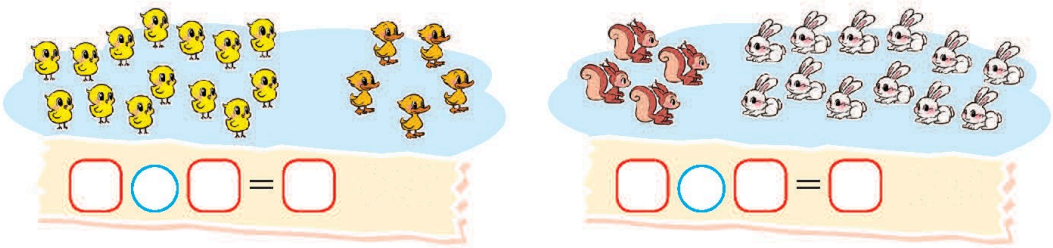
1. Who picked more? How many more?

2. How many fewer ✿ than ✿?

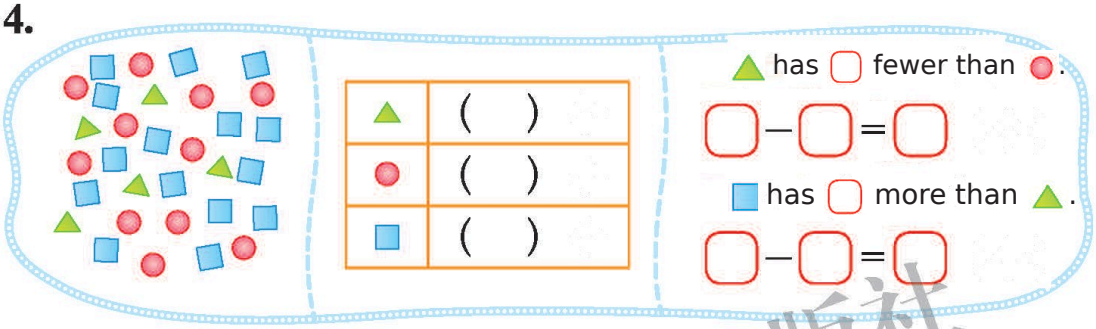


□ ○ □ = □

3. Which group has more? How many more?

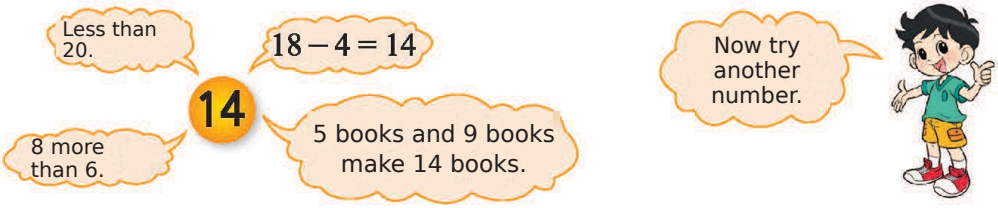


4.



5. Deliver the letters.

6. Talk about it.



7. How many fewer teachers than students?

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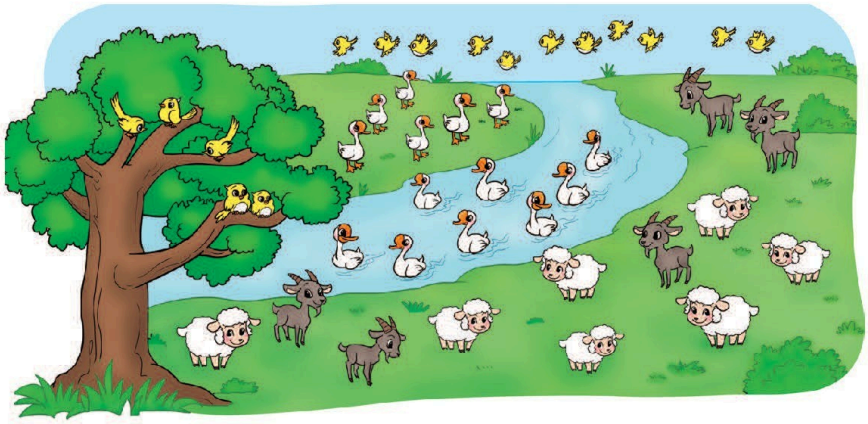
Beautiful Fields



Talk about it: what maths information is in the picture? Fill in.

Sky: ☐ , Tree: ☐

Bank: ☐ , River: ☐

Ground: ☐ , ☐

How many fewer birds are in the tree than in the sky?

☐ ○ ☐ = ☐

Answer: ___ fewer birds in the tree than in the sky.

What problem does Taoqi's number sentence solve?

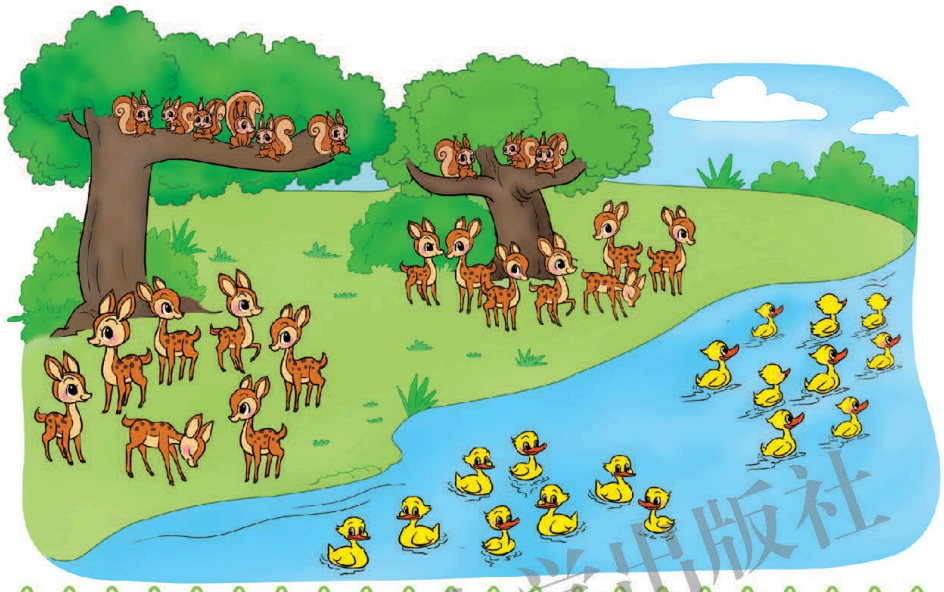
8+6=

Then ask another maths question and try to answer it.

☐ ○ ☐ = ☐

Practice

1.



(1) Fill in.

There are ☐ squirrels, ☐ deer and ☐ ducks.

(2) How many more deer than squirrels?

(3) What problem does this number sentence solve?

$$17-9=\square$$

(4) Ask another maths question and try to answer it.

2.

12 in total

? pencils

☐ ○ ☐ = ☐

? books

11 in total

☐ ○ ☐ = ☐

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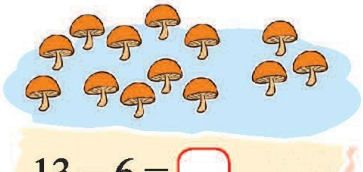
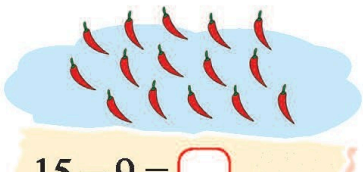
Exercise 1

1. Circle and calculate.



$13-6=\square$



$15-9=\square$

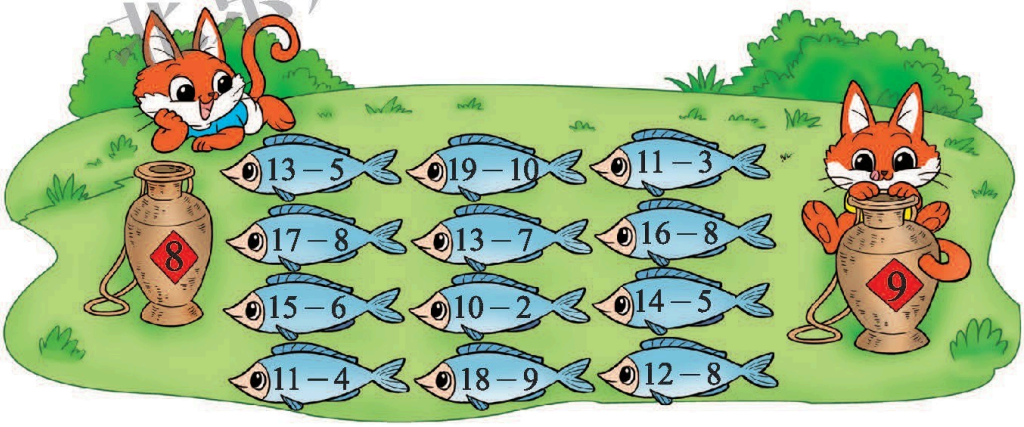
2. Draw and fill in.

$18-9=\square$

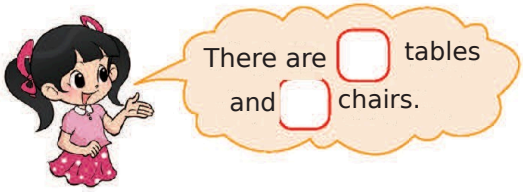
3.

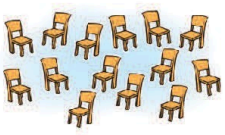
$15-7=$	$16-7=$	$12-6=$	$17-4=$
$15-8=$	$16-8=$	$16-4=$	$14-7=$
$15-9=$	$16-9=$	$16-6=$	$14-6=$

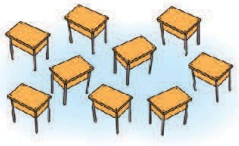
4. Who caught more?



5. 1 chair goes with 1 table. How many more tables are needed?







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6. Write number sentences.

14		
5		9

5 + 9 = ☐
9 + 5 = ☐
14 − 5 = ☐
14 − 9 = ☐

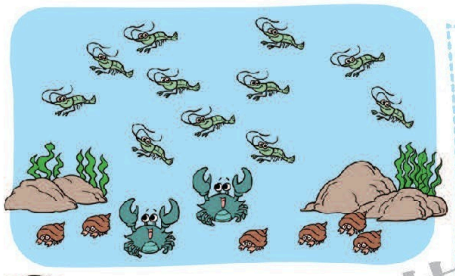
8		
4		12

☐ + ☐ = ☐
☐ + ☐ = ☐
☐ − ☐ = ☐
☐ − ☐ = ☐

6		
7		13

☐ + ☐ = ☐
☐ + ☐ = ☐
☐ − ☐ = ☐
☐ − ☐ = ☐

7.



(1) : ☐ , : ☐ ,
: ☐ .

(2) vs more.
☐ ◯ ☐ = ☐

(3) and in all ☐
☐ ◯ ☐ = ☐

Ask another maths question.

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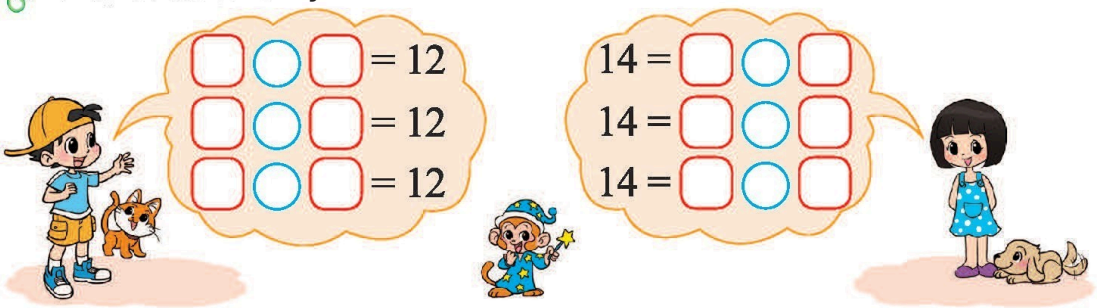
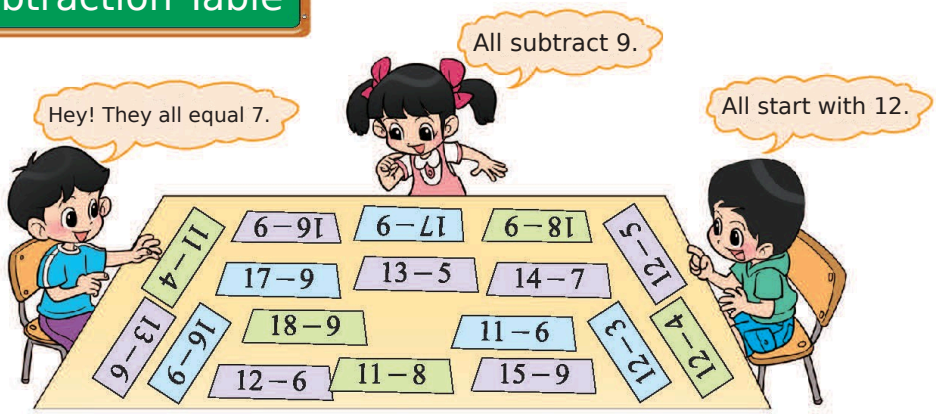
8. Ask two maths questions and try to answer them.

9. See who can say the most.



Subtraction Table

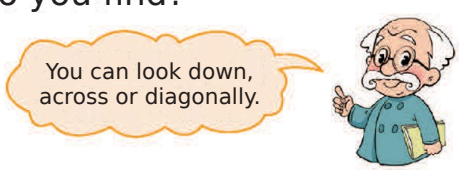


Find and sort number sentences like the children above.

Complete the subtraction table.

10−1	10−2	10−3			10−6			10−9
11−2		11−4	11−5			11−8		
12−3	12−4		12−6	12−7				
13−4	13−5			13−8	13−9			
14−5			14−8	14−9				
15−6		15−8						
	16−8	16−9						
17−8								
18−9								

Discuss: what pattern do you find?



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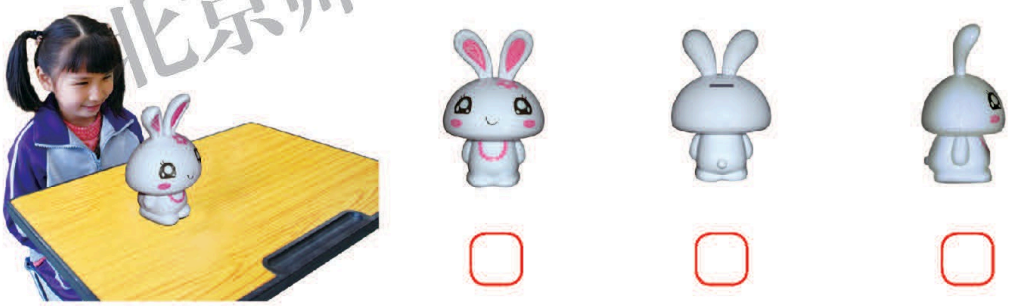
Unit 2 Observing Objects

Look (1)

Look from different directions and say what you see.

Which picture does Xiaoxia see? (Draw "✓")



Work in groups. Look and talk with your partners.

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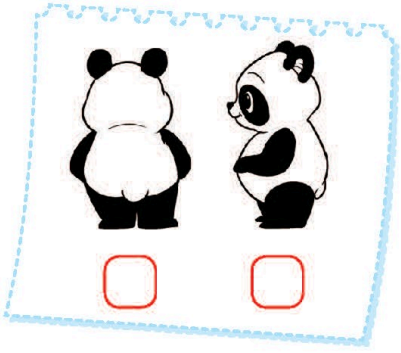
Practice

1. Which picture below does Taoqi see? (Draw "✓")



2. What do they each see? Match.

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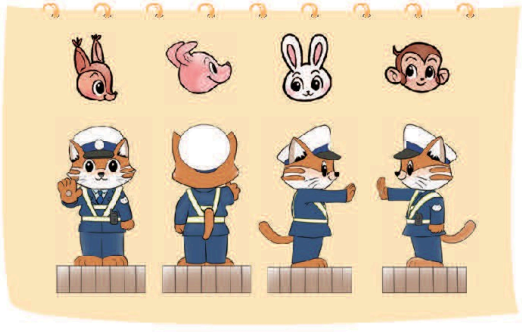
3.

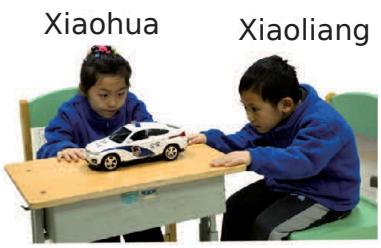
$12-7=$	$16-8=$	$13-5=$	$7+6=$
$11-4=$	$9+3=$	$17-9=$	$14-6=$

4. Who sees each picture below? Match.



Look (2)

Observe from different directions and say what you see.



Which picture below do they each see? Match.

Xiaoming Xiaohua Xiaoliang

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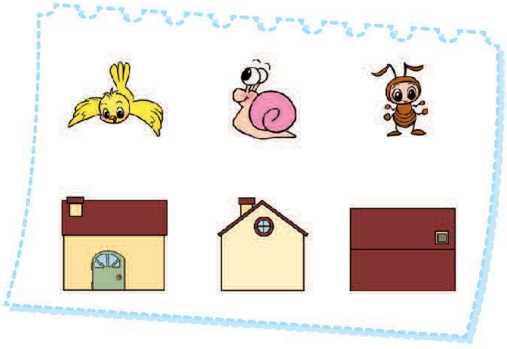
Which picture shows the back of the schoolbag? (Draw "✓")
Which picture shows the side of the schoolbag? (Draw "○")

Practice

1. What do they each see? Match.



2. Look and talk. Which picture below does Taoqi see? Which picture does Xiaoxiao see?

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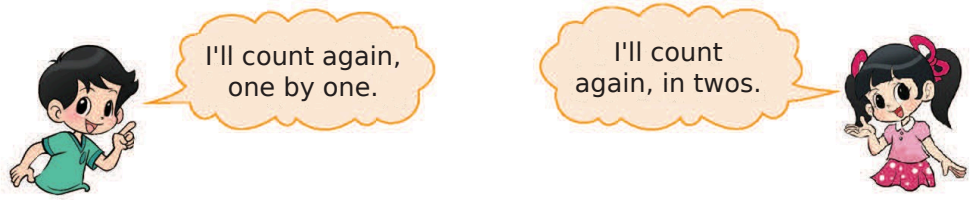
3. Match and talk.

Unit 3 Numbers in Life

Count Peanuts

How many peanuts do Taoqi and Xiaoxiao each have? Count one by one.

Is the count right?



Count again another way.

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Practice

1. Count out 36 peanuts one by one.

   Count out 80 peanuts in fives.

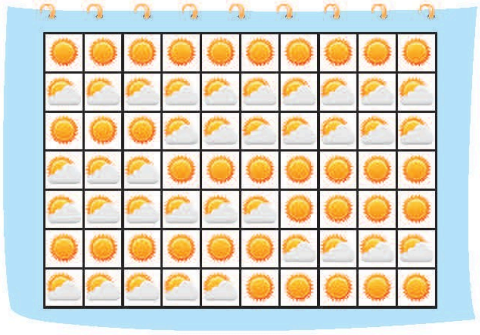
2. Say what comes next.

3. Say the next five numbers after each one.

   twenty-three, fifty-six, seventy-seven, eighty-nine, ninety-five

4. Circle and count.

5. How many and are there? Say how you counted.



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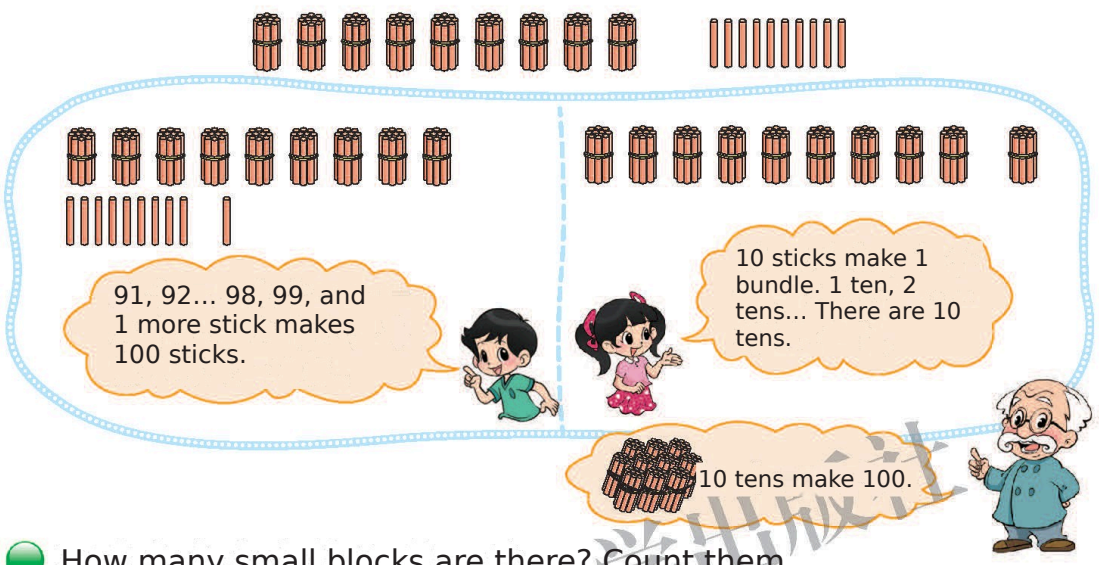
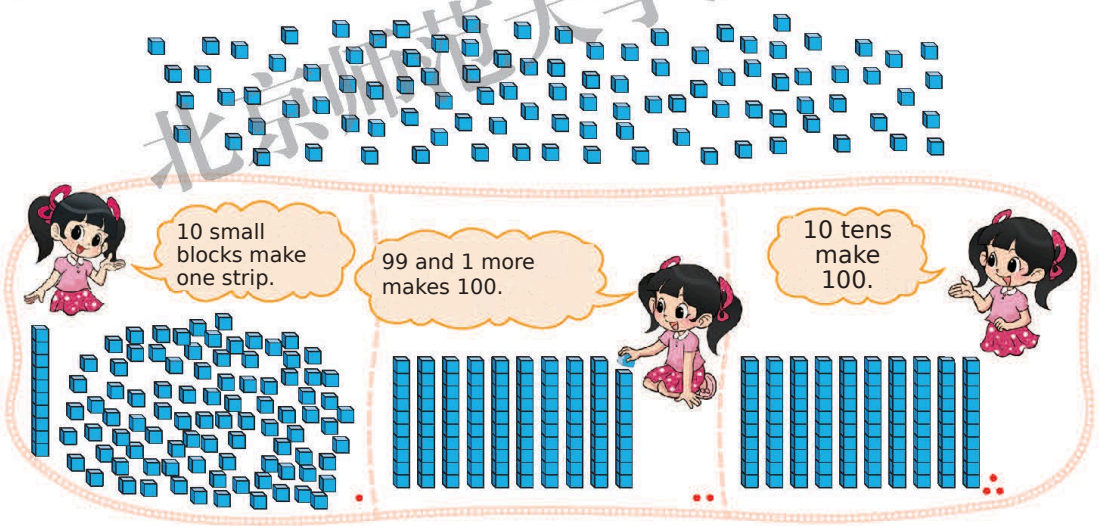
Count

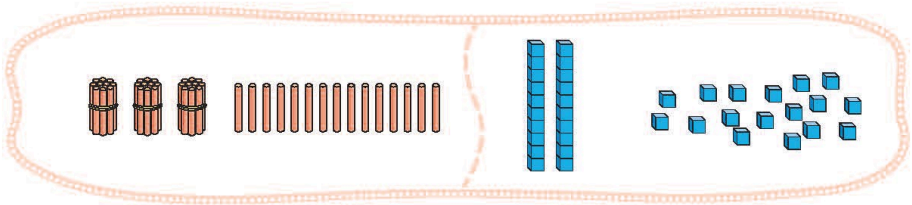
How many sticks? Count them and read the number.



How many small blocks are there? Count them.



Circle and count.



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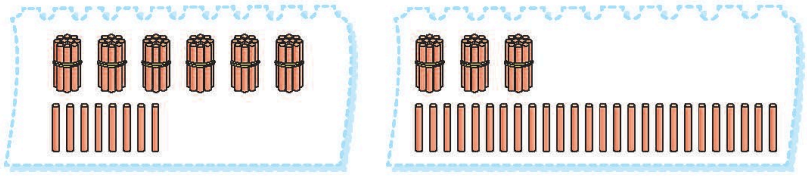
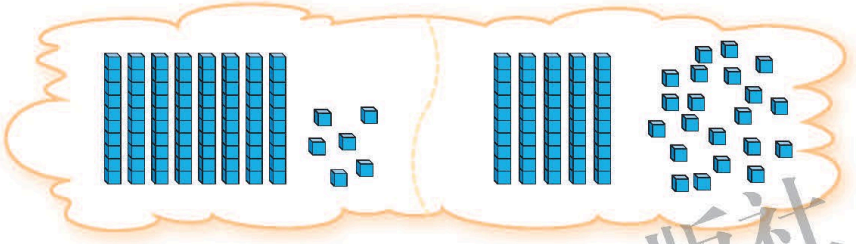
Practice

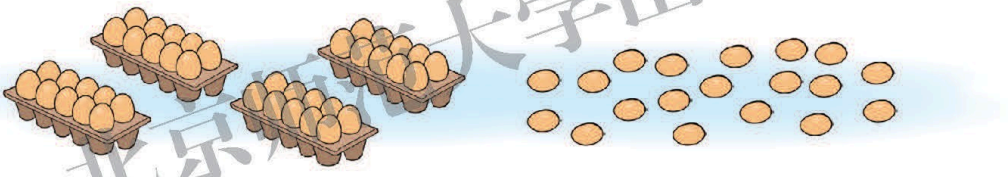
1. Circle and count. How many sticks are there?



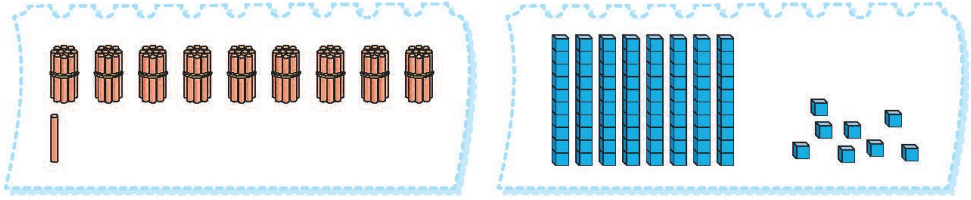
2. Circle and count. How many small blocks are there?



3. Circle and count. How many eggs are there?



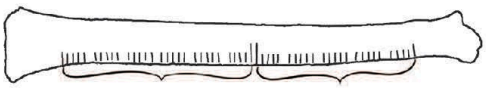
4. Count. How many more are needed to make one hundred?



5. Read and discuss.

In ancient times, people counted by carving deep marks on stone slabs or animal bones.

People found this wolf bone. It is about 30,000 years old. Can you work out how ancient people counted?



Group 2: 30 marks Group 1: 25 marks

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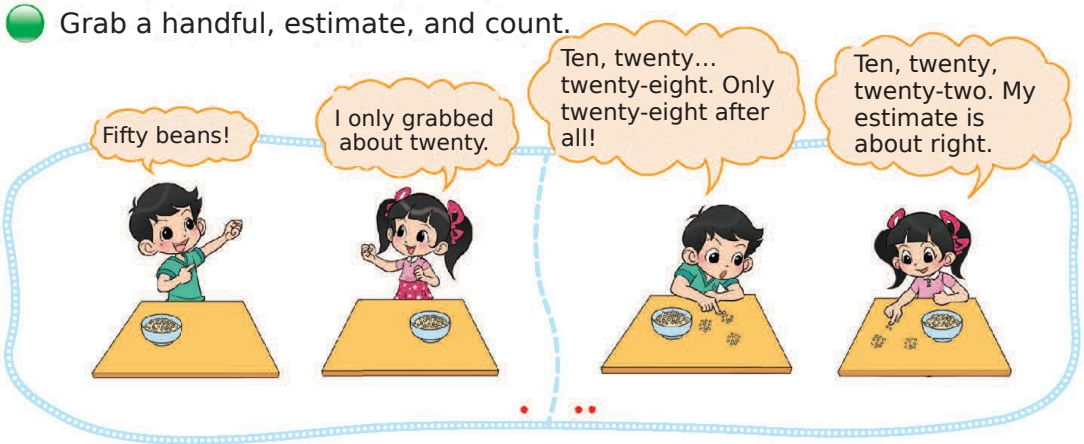
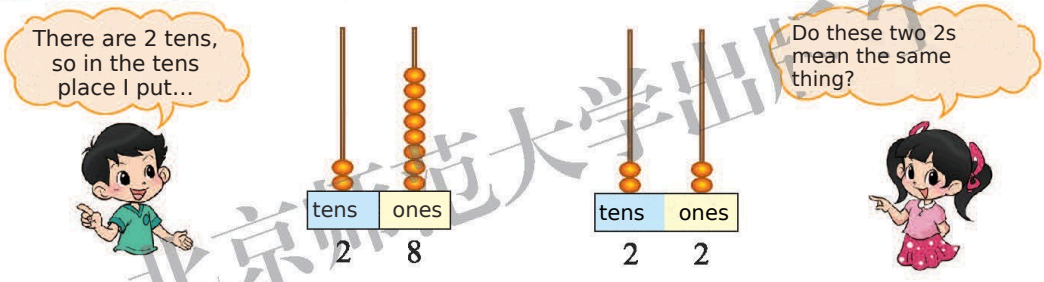
Count Beans

Grab a handful, estimate, and count.



How do you show it and write it?



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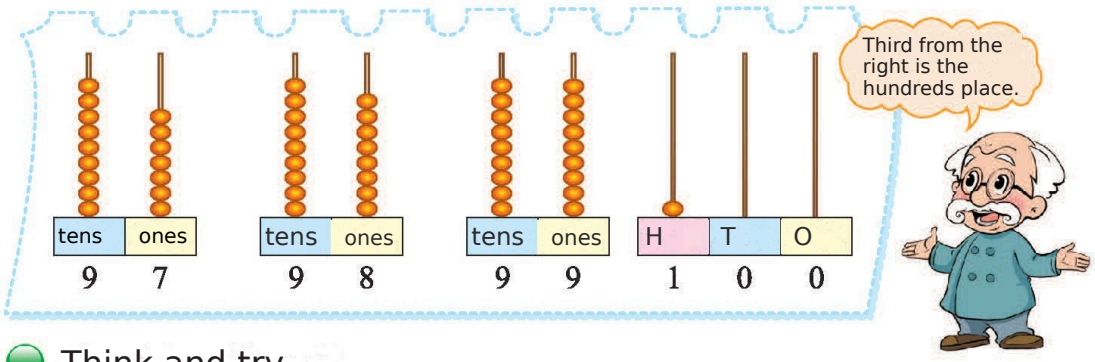
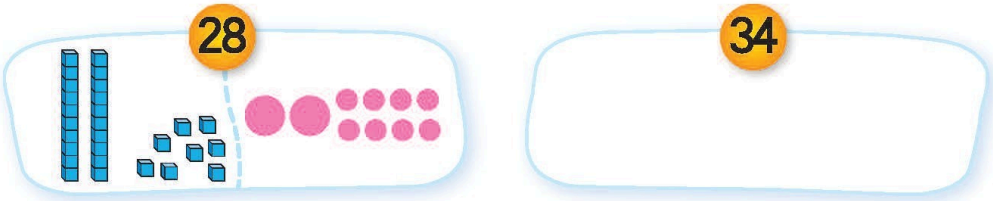
Show and read the numbers.

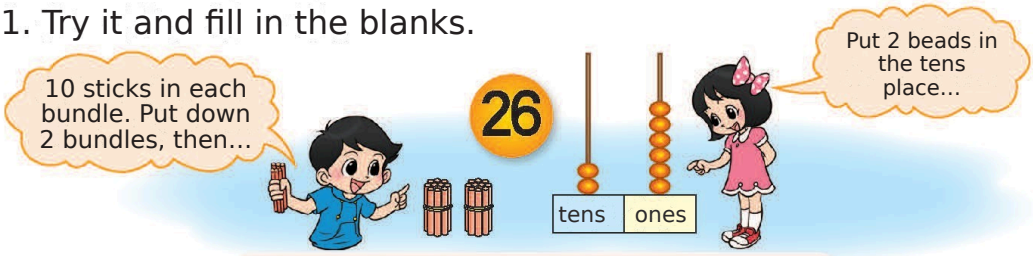


Think and try.



Practice

1. Try it and fill in the blanks.



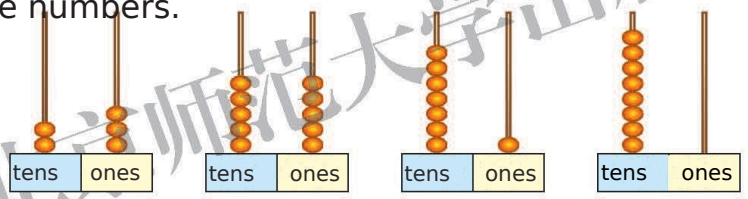
( ) tens and ( ) ones make 26.

2.

( ) tens and ( ) ones make ( ).

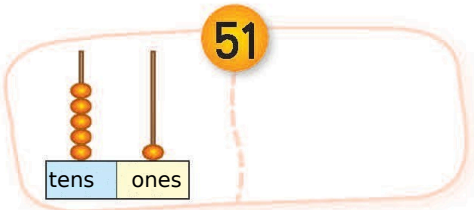
( ) tens and ( ) ones make ( ).

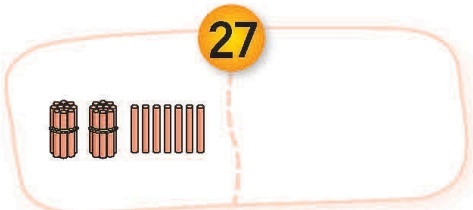
3. Write the numbers.



( ) ( ) ( ) ( )

4. Think and try.





5. Find 100 in daily life.

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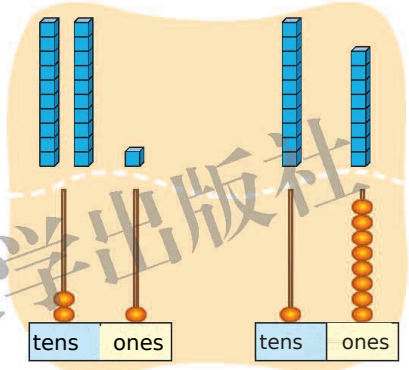
Who Has More?

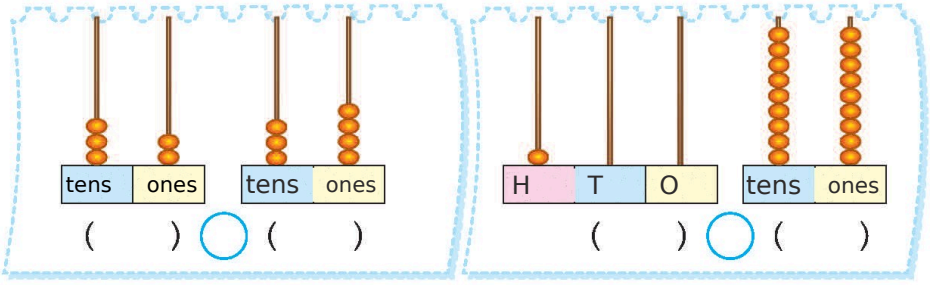
Who has more red fruits?

Try it and fill in the blanks.



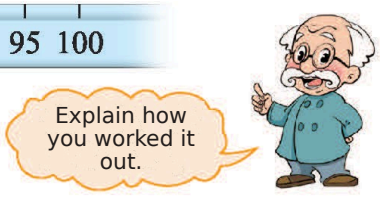
□ > □

Write and compare.



Look and fill in.

45 < □ 90 > □



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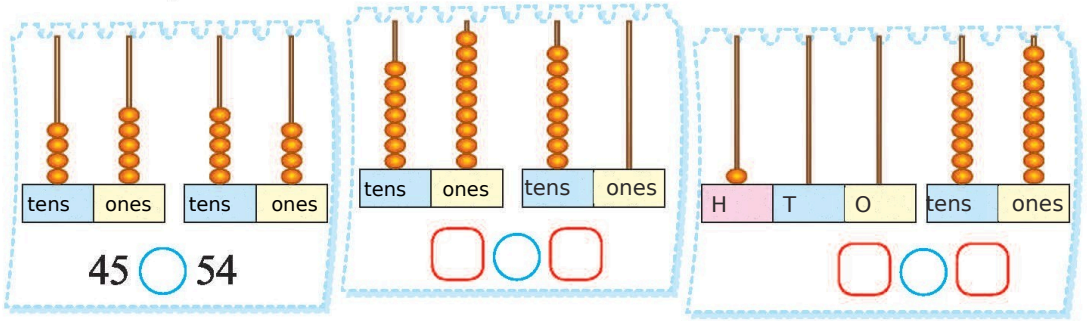
Practice

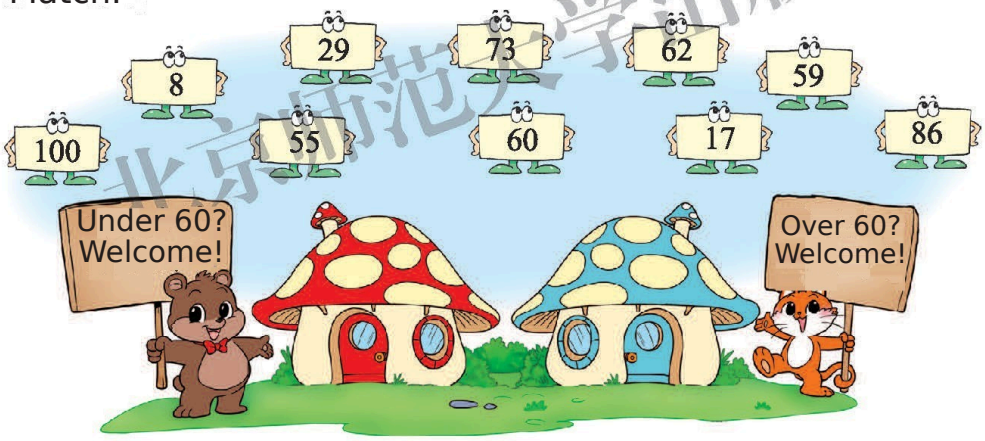
1. Compare.



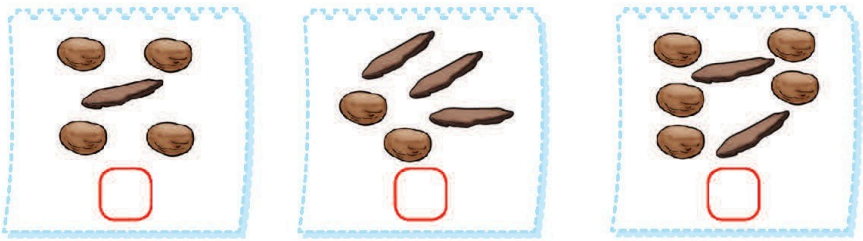
2. Fill in.

15 < ☐ ☐ > 89 ☐ < 30 80 > ☐

3. Match.



4. means 10, means 1. Tick the numbers greater than 30.



5. How many Grade 1 classes are in your school? Which class has the most students? Which has the fewest?

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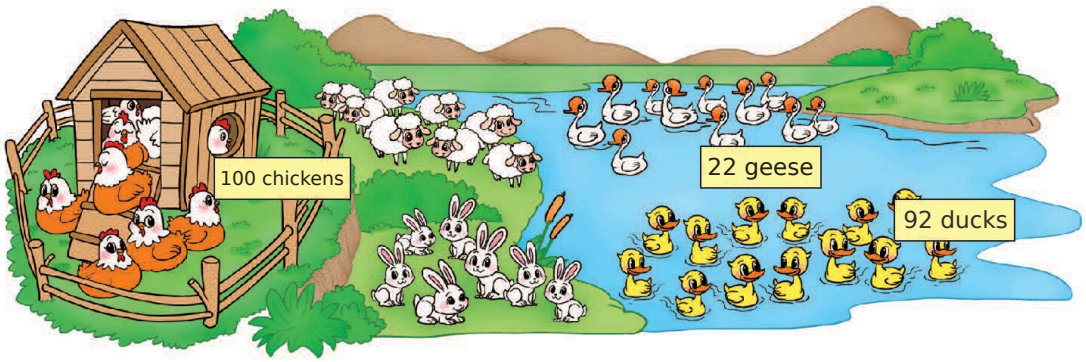
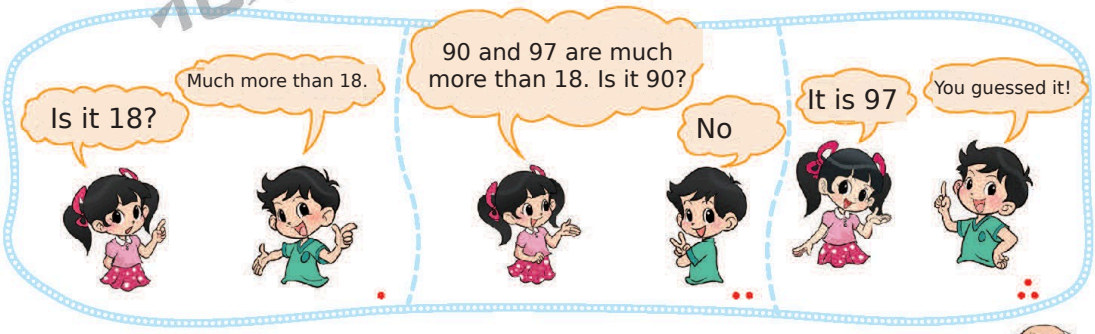
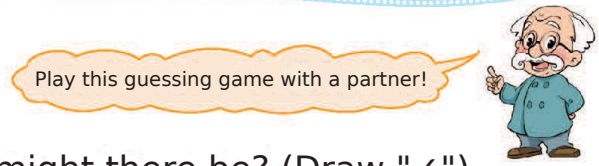
A Little Farm



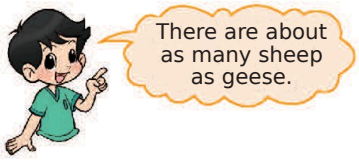
Talk about it. Which are there more of? Which fewer?

One of the four numbers 18, 26, 90 and 97 is the number of rabbits. Guess which.





Think: how many sheep might there be? (Draw "✓")



70 sheep	26 sheep	3 sheep

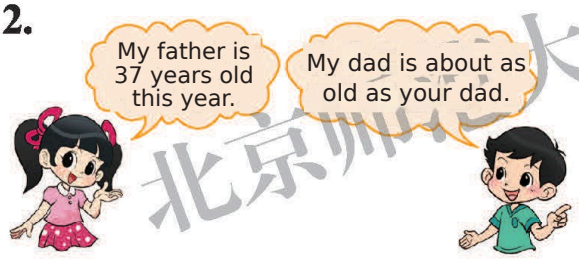
Practice

1. 86 people are running.

How many might do the long jump? (Draw "○")
How many might skip rope? (Draw "✓")

88 people	12 people	76 people

2.



How old might Taoqi's father be this year? (Draw "✓")

39 years	50 years	28 years

3.

Draw "✓" under the answer you think is right.

30	55	90

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A Hundred Chart

Look and fill in.

1									10
	12							19	
		23					28		
			34			37			
				45	46				
				55	56				
			64			67			
		73					78		
	82							89	
91									100

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Talk about it. What do you notice?

Think and fill in.

12	13	14
	23	24
		34

58		60
78		80

	67	

Practice

1. Colour in the completed hundred chart, look at it, and tell your partner what you notice.

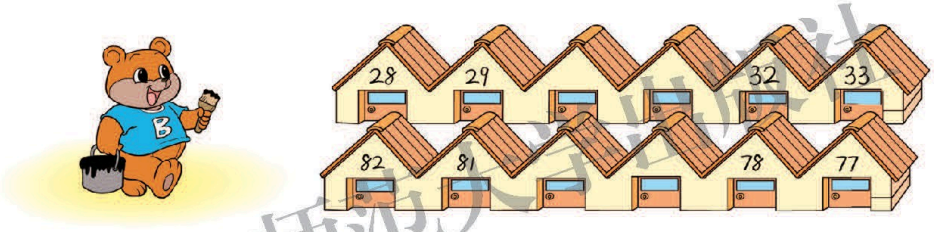
(1) Colour green the numbers whose ones digit is "0".

(2) Colour blue the numbers whose ones digit is "7".

(3) Colour yellow the numbers whose two digits are the same.

(4) Colour red where the ones digit is 1 less than the tens digit.

2. Little Bear writes the house numbers.



3. Look and fill in.

4. Think and fill in.

27	28	
	38	
		49

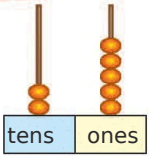
	31	
40		42
	51	

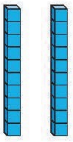
	85	

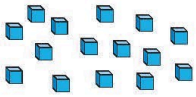
Exercise 2

1. Fill in.

___ tens and ___ ones make ___. ___ tens and ___ ones make ___.

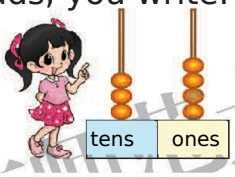




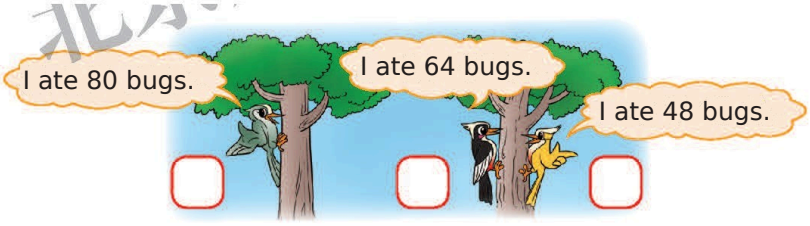


___ tens and ___ ones make ___. ___ tens and ___ ones make ___.

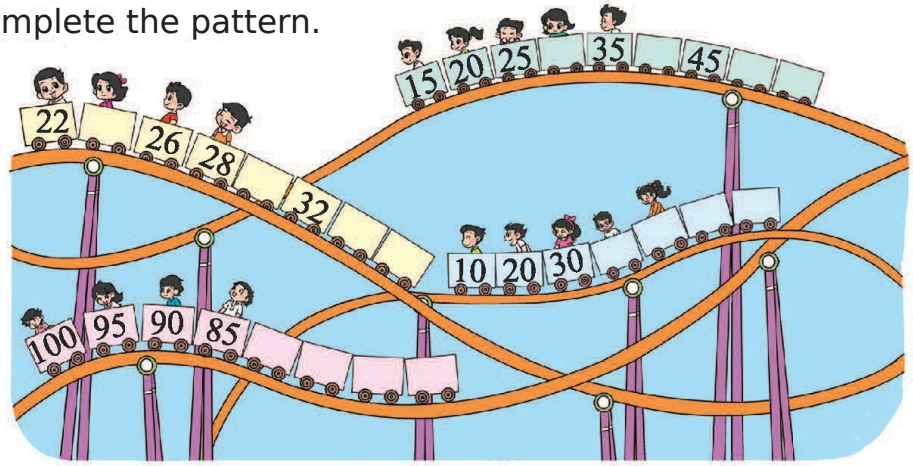
2. I move the beads, you write.



3. Who ate the most bugs? (Draw "○") The fewest? (Draw "✓")



4. Complete the pattern.



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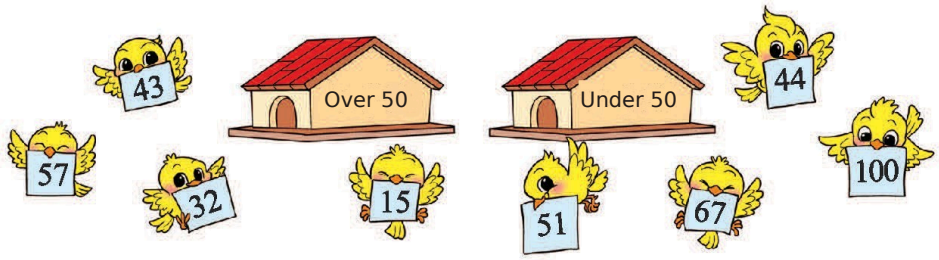
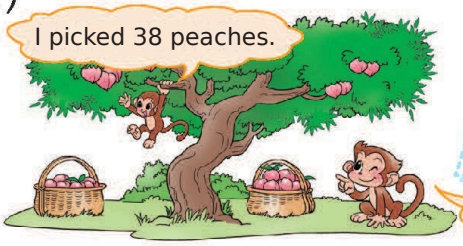
5. Birds Go Home (Match)



6. Draw "○" under the answer you think is right.

(1)

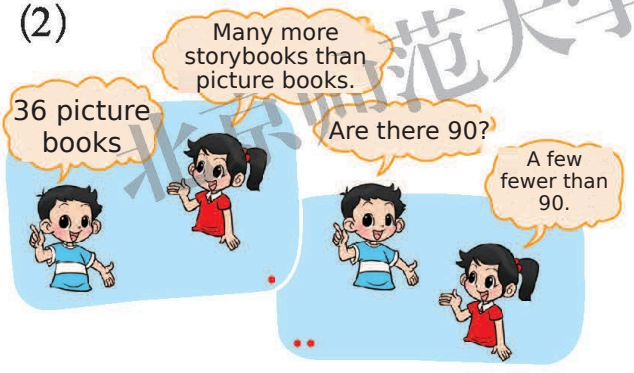


How many peaches might the big monkey have picked?

96	42	35

I picked a few more than you.

(2)



How many storybooks might there be?

85 books	99 books	40 books

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7. What numbers can 4 beads show on a counting frame? Draw and write.

tens	ones	tens	ones	tens	ones	tens	ones	tens	ones
(13)		( )		( )		( )		( )	

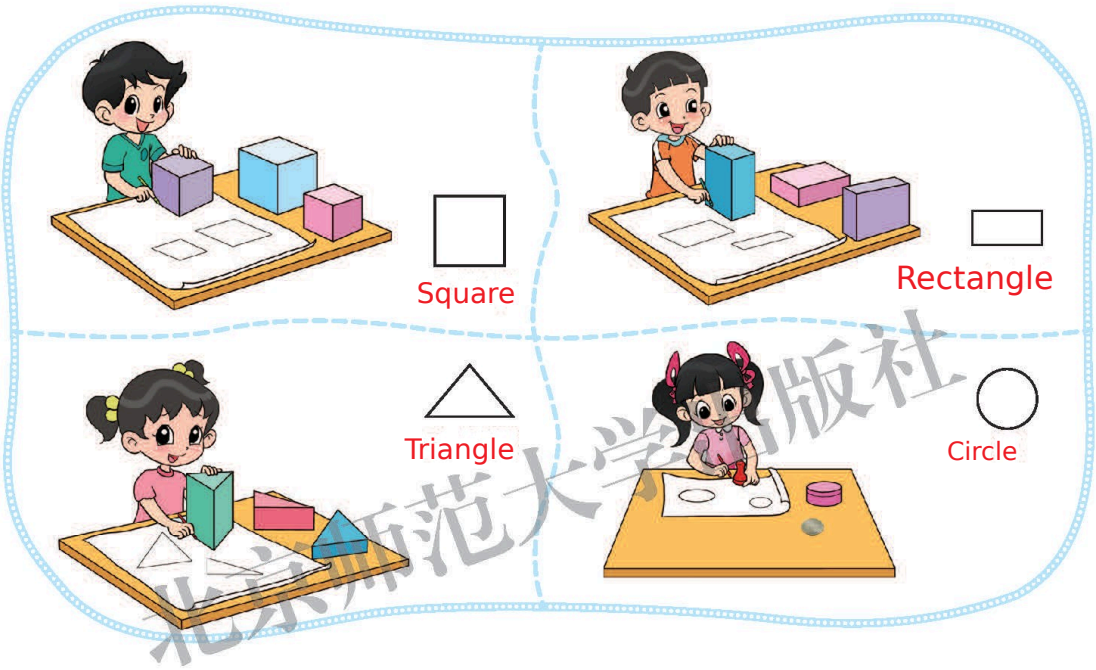
8. 2 5 8 Use two of these cards to make a two-digit number.

What is the largest? What is the smallest?

Unit 4 Interesting Shapes

Learn Shapes

Try it. What shapes can you get? Name them.



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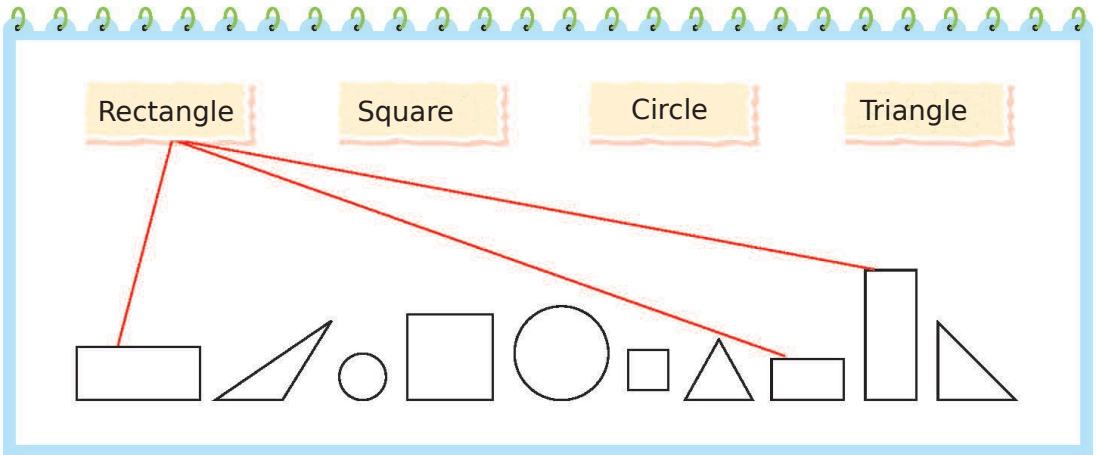
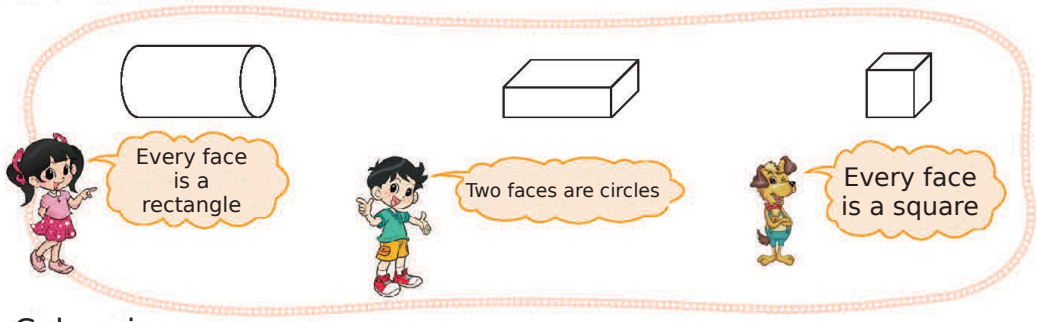
Sort and match.



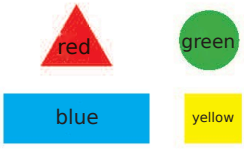
Talk about it: where else have you seen these shapes?

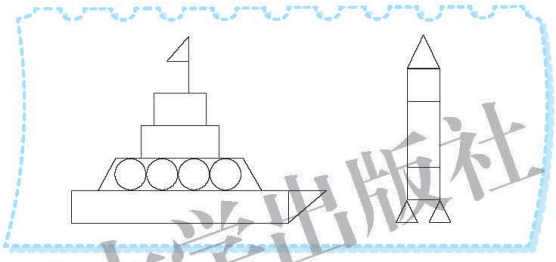
Practice

1. Match

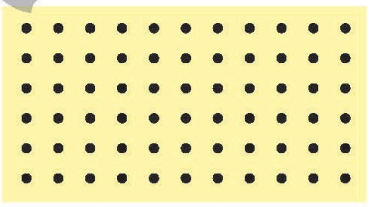


2. Colour in.

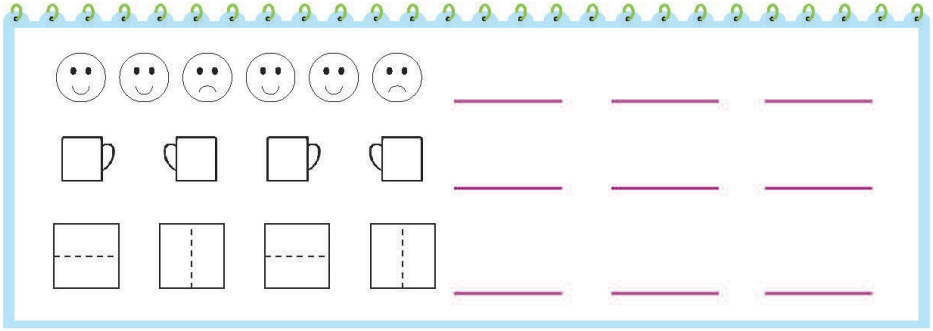




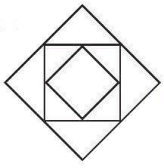
3. On the dot grid, draw a rectangle, a square and a triangle.



4. Think about it, then continue the pattern.



5. Count.



The figure has __ △ and __ □.

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Hands-on (1)

Fold the paper square in Figure 1 of the appendix into two equal parts. How do you fold it?

Cut and compare.

Fold the paper rectangle, triangle and circle in Figure 2 on the appendix page each into two equal parts. How do you fold them? Cut and compare.

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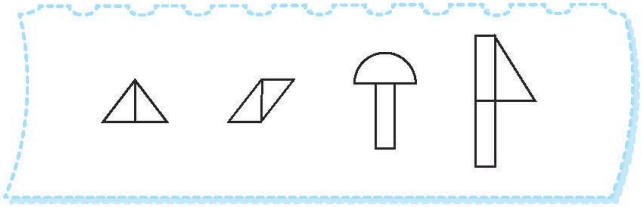
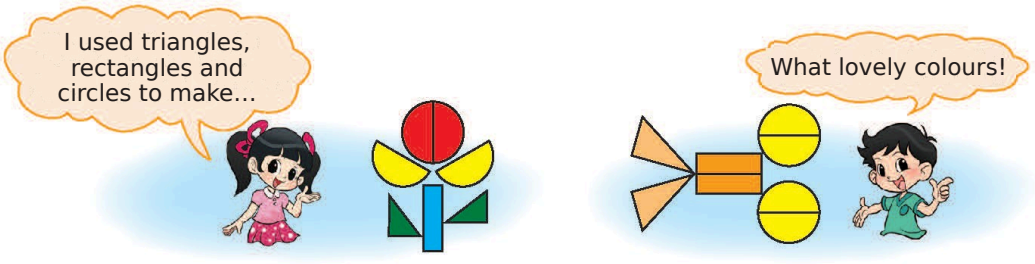
Use your cut-out shapes to make these pictures.

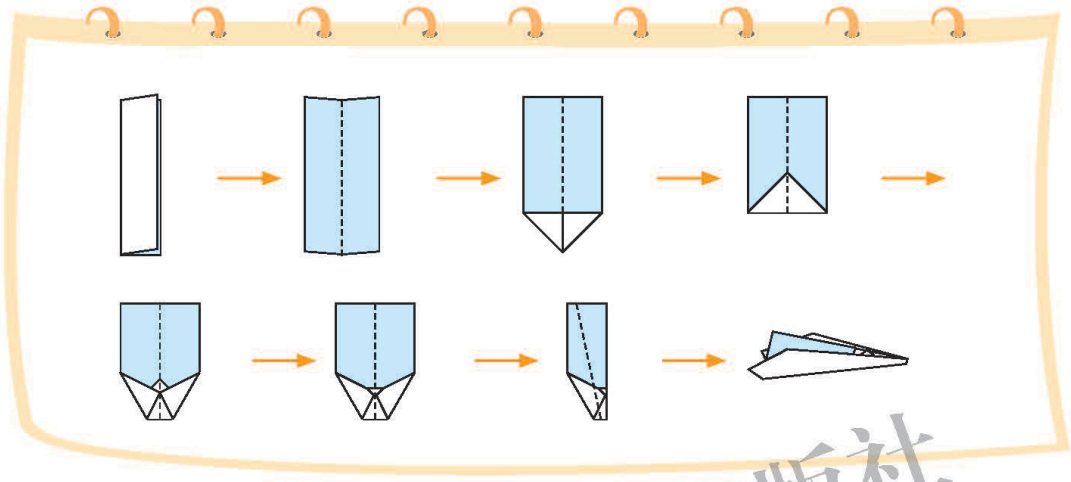


Can you work with a partner to make other shapes?

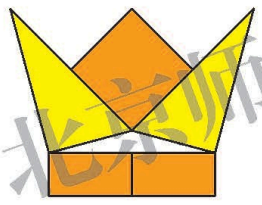


Practice

1. Fold a paper aeroplane.



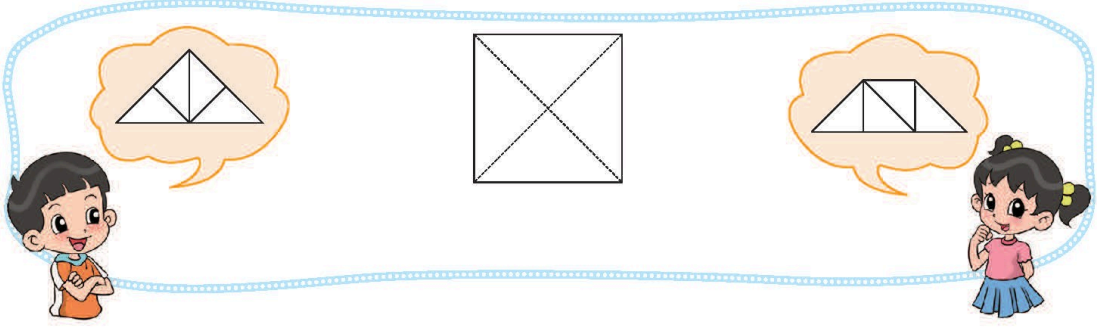
2. What does this look like? Cut out Figure 3 in the appendix and make it.



Can you make other interesting shapes?

3. Can you fold a rectangular piece of paper to make a square? Try it.

4. Cut a square piece of paper into four triangles of the same size. What shapes can you make with these triangles?



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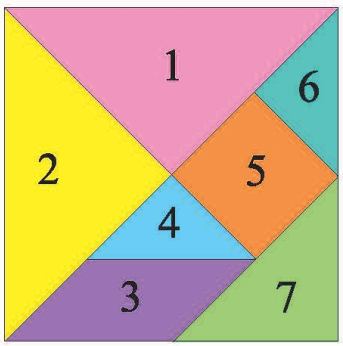
Hands-on (2)

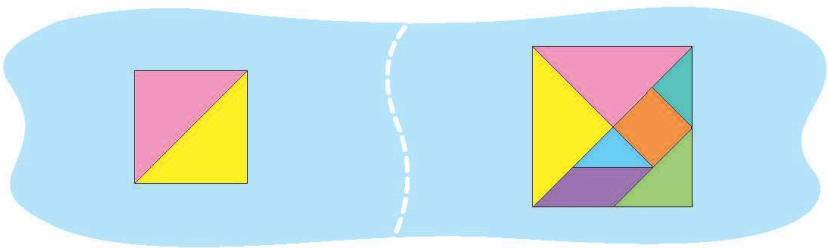
This is a tangram. Shape 3 is a parallelogram.



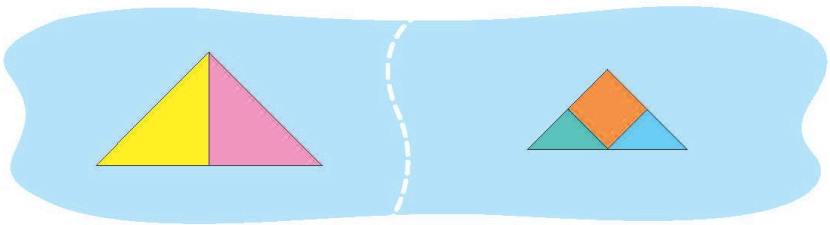
In the Song Dynasty, a man named Huang Bosi invented a table that could be arranged into different shapes depending on how many people were eating: a triangle for 3 people, a square for 4 people... Interesting, isn't it?

Later, this table evolved into a toy that is very clever and fun. People call it the "tangram".

- Talk about it and fill in. The tangram has ___ kinds of shapes, including ___ triangles. Shapes 1 and ___, and shapes 4 and ___, are exactly the same.
- Try to make a square.



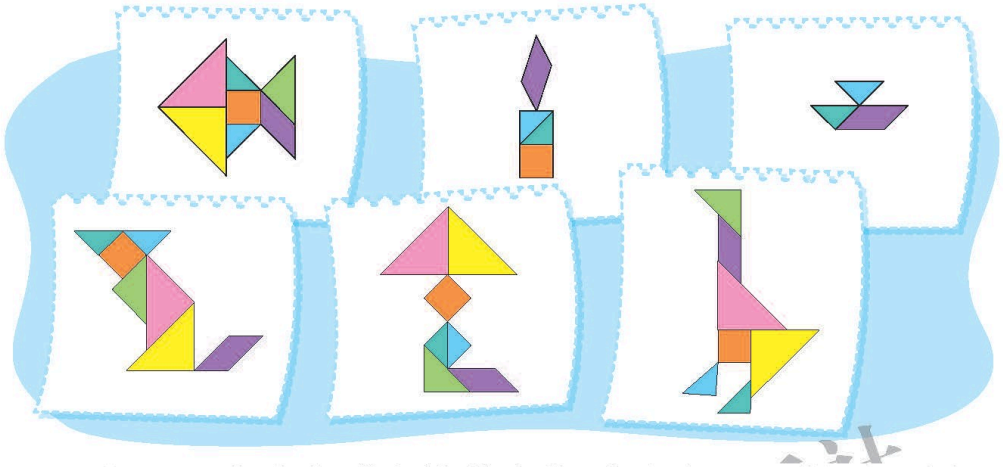
- Try to make a triangle.



Can you find other ways to make it?

Practice

1. Make these pictures. What does each one look like?



2. These pictures tell the story "Waiting by a Tree for a Hare". Make them.

3.

$8+9=$	$14-8=$	$15-7=$	$12-4=$
$11-3=$	$13-6=$	$17-5=$	$14-2=$

4. Use your imagination to make your favourite picture with the tangram, and show it to your classmates.

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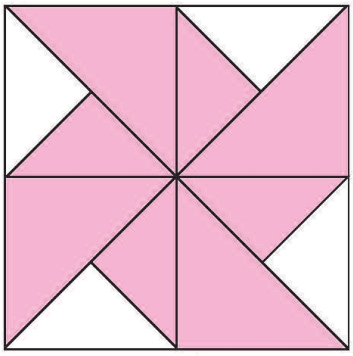
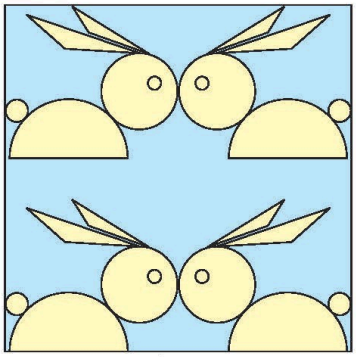
Hands-on (3)

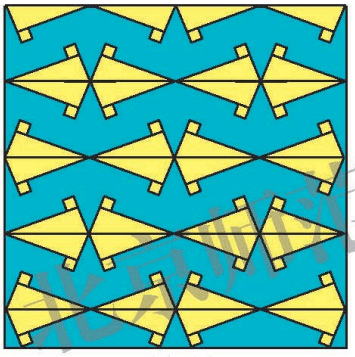
Enjoy the patterns. Find the shapes you know.



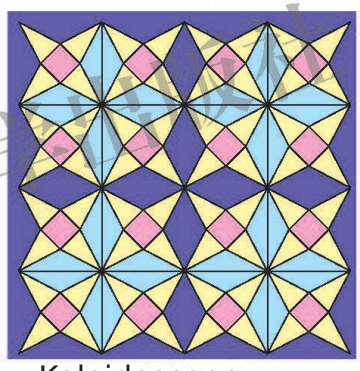
Pinwheel



Rabbits



Goldfish



Kaleidoscope

The pinwheel pattern is made of 8 triangles...

The kaleidoscope pattern uses triangles and squares...

Design a pattern with a partner. Say which shapes you used.

With a circle and triangles I made a...

This pinwheel uses 4 identical triangles.

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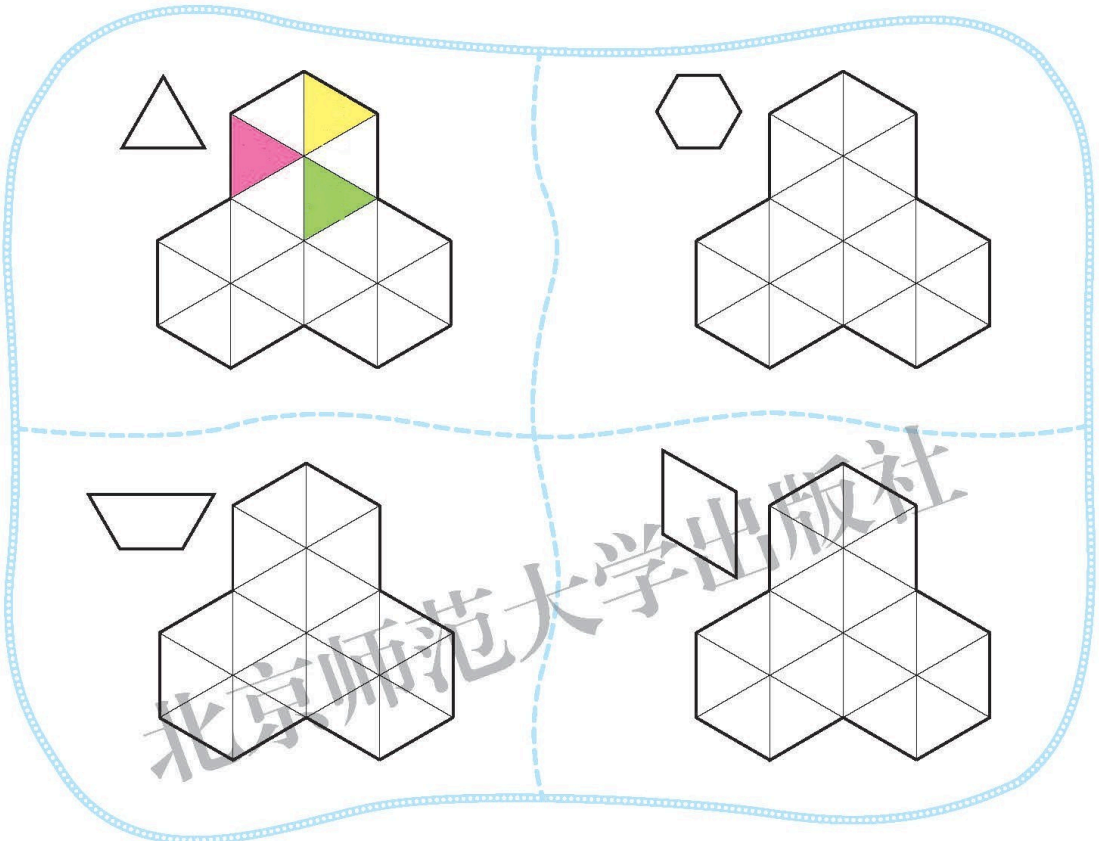
Practice

1. Colour the shapes that match the one in the top left corner.



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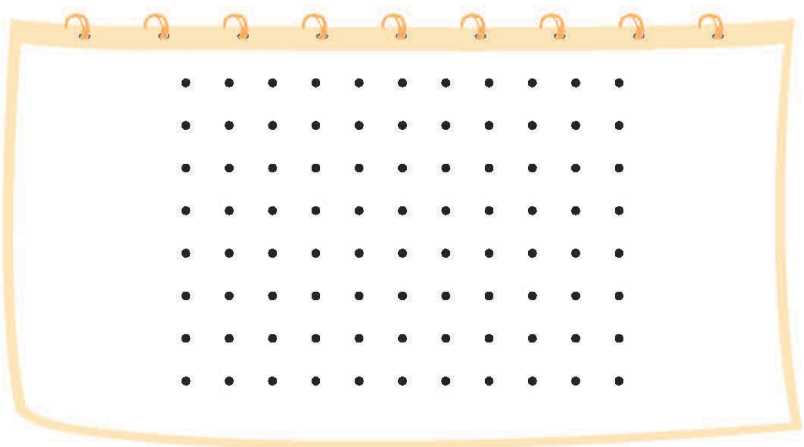
2. Design a pattern on the dot grid with shapes you know. Which shapes did you use?



Review

What I Learned

Compare

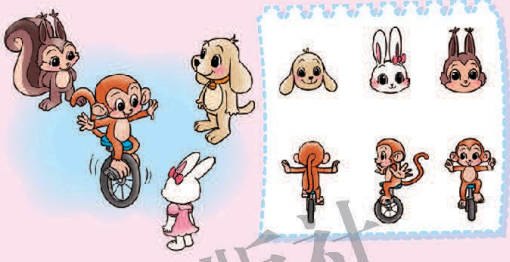
15 plates

6 plates

How many fewer plates does the cat have than the elephant? Draw a picture to explain.

Work it out and tell your partner how you did it.

Match



Choose your favourite toy, look at it from different directions, and talk about it.

Talk About It

10 tens

1 more than 99

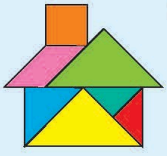
100

2 fifties

100 ones

Find a number greater than 20 and describe it like this.

Make It



Can you make a rectangle with the tangram? In how many ways?

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My Learning Journey

It took me a long time to make a square with 5 tangram pieces.

When I got stuck, laying out sticks helped.

I only knew one way to compare numbers. Group discussion taught me another.

The number I remember most from the survey is...

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My Questions

Why does the tangram have no rectangles?

Why are all the shapes that can stand up solid shapes? Can we find a way to make flat shapes stand up?

Numbers in life never end. Can we learn them all?

Mum says there are also negative numbers. Is a negative number smaller than 0? How can that be? What use are such numbers? I just can't write a number smaller than 0.

Review and Apply

1. Work these out with sticks or a counting frame, and say how you did it.

$8+5=$ $11-6=$ $17-8=$ $13-5=$

2. Draw and fill in.

$16-8=\square$

3.

$10-8=$	$8+4=$	$12-4=$	$14-6=$
$15-6=$	$5+7=$	$17-9=$	$3+8=$
$11-3=$	$15-8=$	$14-9=$	$11-2=$

4.

$3+9=\square$
$12-3=\square$
$12-9=\square$

$7+6=\square$
$13-\square=\square$
$13-\square=\square$

$14-5=\square$
$15-4=\square$

$18-9=\square$
$19-8=\square$

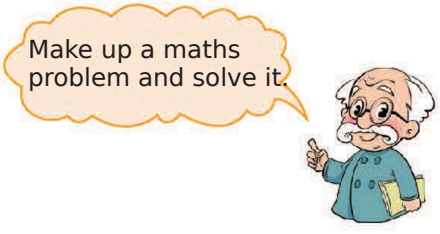
5.

16 cherries ? cherries
ate 7 cherries
$\square\bigcirc\square=\square$

? cherries 9 cherries
ate 7 cherries
$\square\bigcirc\square=\square$

6. Compare lifespans.

15 years	9 years



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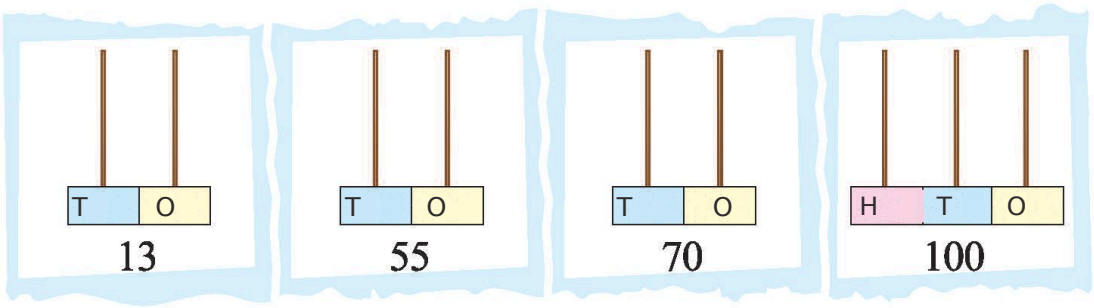
7. Draw and move the beads.



8. What do they each see? Match.

9. Can you make a pretty picture with the shapes you have learned? If it is hard, you can copy one of the pictures below.

10. Kicking the shuttlecock

(1) How many kicks did Xiaowen and Xiaoli make in all?
(2) How many fewer kicks did Xiaowen make than Xiaoli?
(3) How many more kicks does Xiaojun need to match Xiaoli?

11. Xiaoxiao uses the picture below to show 23. Can you show 32 in the same way?

12.

How many bottles might Taoqi have collected? (Tick one.)

45	80	30

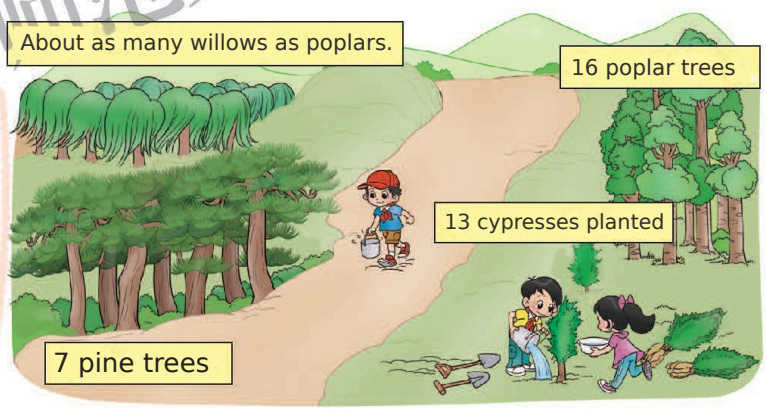
13. Fill in each ◯ with >, < or =.

36 ◯ 63	65 ◯ 60	45 ◯ 45	89 ◯ 100
19 ◯ 91	100 ◯ 92	42 ◯ 39	31 ◯ 29

14.

How many willow trees might there be? (Tick one.)

42	81	15



(1) How many more cypress trees than pine trees?

(2) Make up a maths problem and solve it.

(3) Add the missing information and solve: Are there more pine trees or locust trees? How many more?

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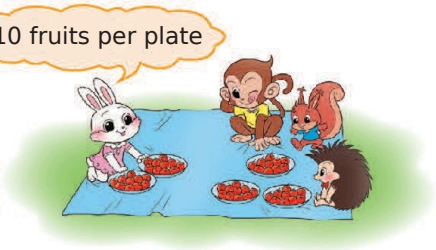
Unit 5 Addition and Subtraction (2)

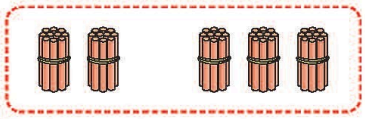
Bunny's Party

How many fruits are there in total?



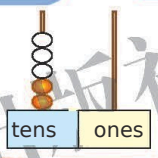
□ ○ □ = □

Count in tens, 30, 40, 50.



2 tens and 3 tens make 5 tens, which is 50.

$2+3=5$
$20+30=50$

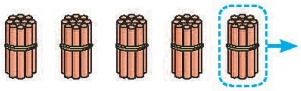


Answer: ____ fruits in total.

How many fruits are left?

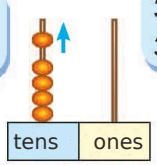
□ ○ □ = □

Count: 50, 40.



5 tens minus 1 ten is 4 tens, which is 40.

$5-1=4$
$50-10=40$



Answer: ____ fruits are left.

Let's learn.

$20 + 30 = 50$

Addend Addend Sum

$50 - 10 = 40$

Minuend Subtrahend Difference

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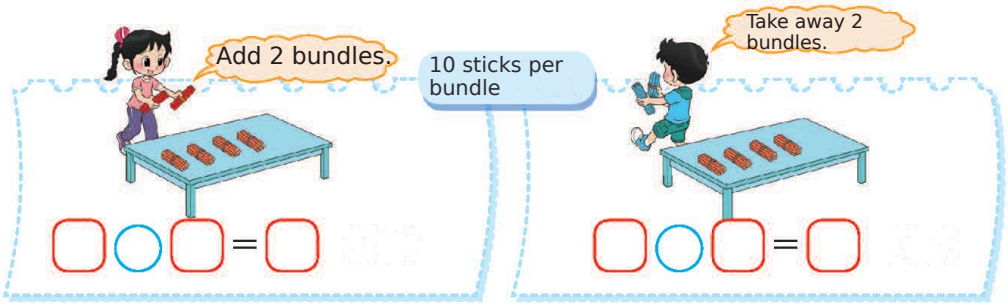
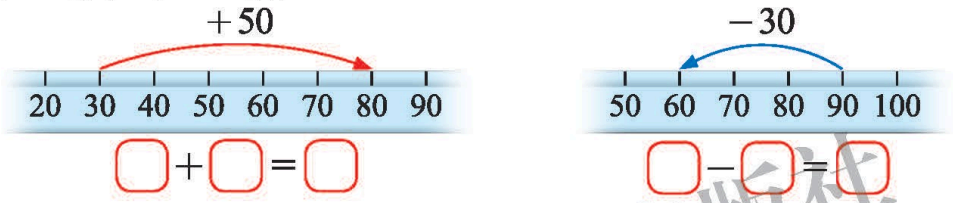
Practice

1.



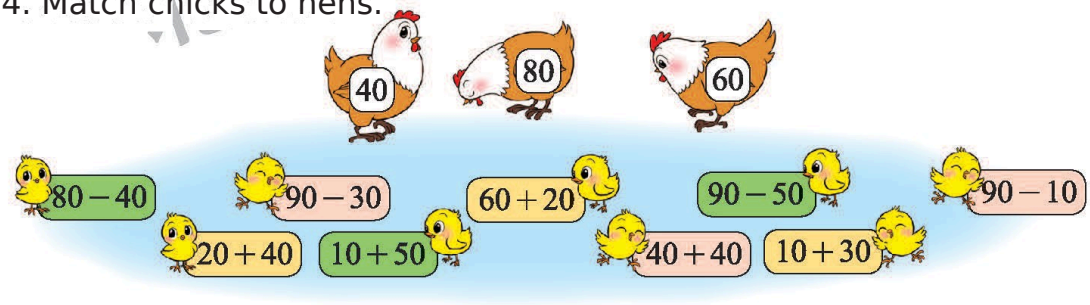
2. Look and fill in.



3.

60 + 30 =	70 + 20 =	50 + 20 =	10 + 30 + 50 =
90 − 20 =	80 − 50 =	60 − 50 =	90 − 50 − 20 =

4. Match chicks to hens.



5. Make up a maths problem and solve it.

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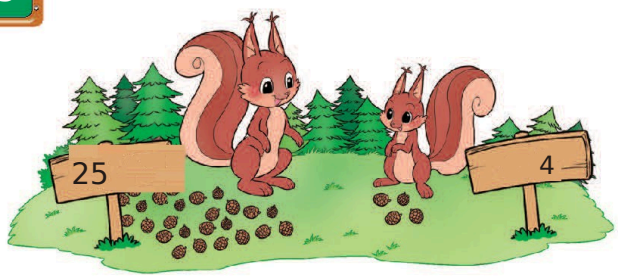
Pine Cones

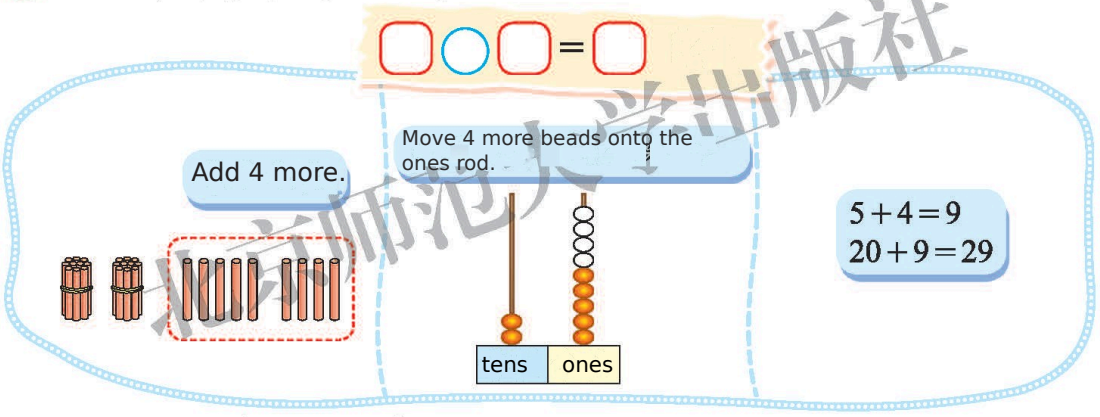


What maths problems can you ask?

How many pine cones were collected in all?

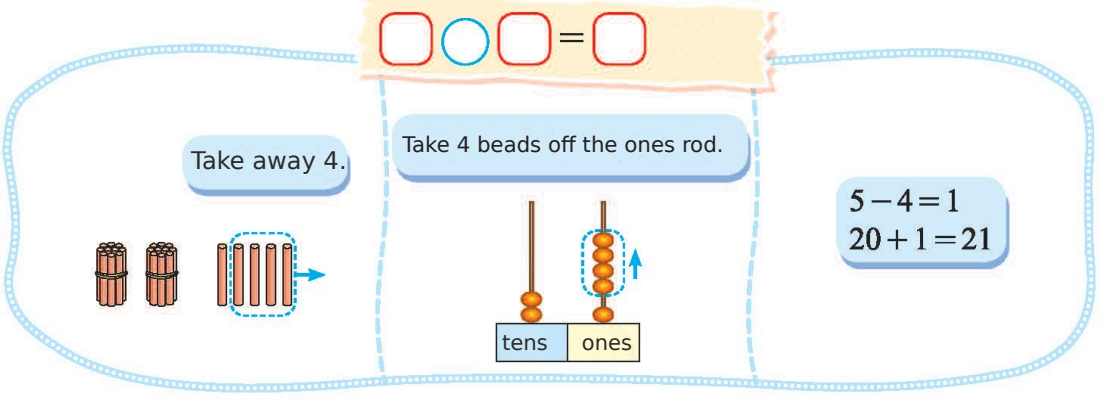
How many more pine cones did Mother Squirrel collect than Little Squirrel?

How many pine cones were collected in all?



Answer: ____ pine cones in all.

How many more pine cones did Mother Squirrel collect than Little Squirrel?



Answer: Mother Squirrel collected ____ more pine cones than Little Squirrel.

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Practice

1. Make up a maths problem and solve it.

2. Work these out on a counting frame.

$32+5=$ $5+74=$ $78-6=$

3.

How many are left?

I ate 9

19 at first

□ ○ □ = □

How many metres longer is the big dinosaur than the small one?

2 m long 25 m long

□ ○ □ = □

4.

$4+65=$	$72+5=$	$67-2=$	$44-4=$
$83+6=$	$26+3=$	$88-5=$	$47-5=$

5. Think and fill in.

$42+3=$□ $62+5=$□ $7+40=$□

$3+42=$□ $5+62=$□ $40+7=$□

Can you write two pairs of number sentences like the ones above?

□ + □ = □ □ + □ = □

□ + □ = □ □ + □ = □

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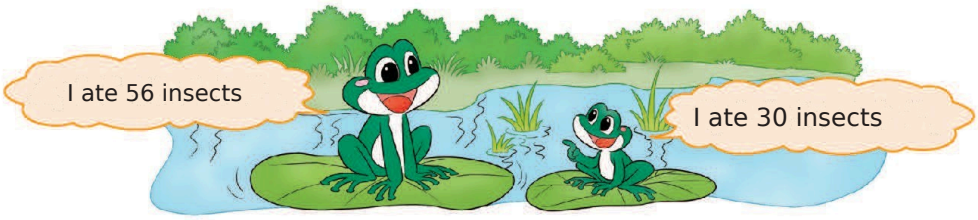
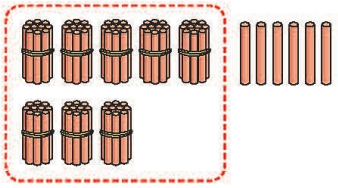
Frogs Eat Insects

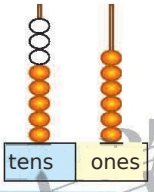


How many insects were eaten in all?

□ ○ □ = □



Move 3 more beads onto the tens rod.

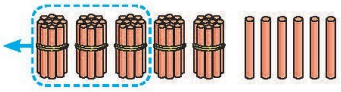


$50+30=80$
$80+6=86$

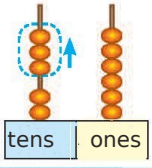
Answer: They ate ____ insects in all.

How many fewer insects did the small frog eat than the big frog?

□ ○ □ = □



Take 3 beads off the tens rod.



$50-30=20$
$20+6=26$

Answer: The small frog ate ____ fewer insects than the big frog.

Calculate and explain.

Add or subtract ones with ones...

$34+20=$ $71-10=$
$34+2=$ $71-1=$

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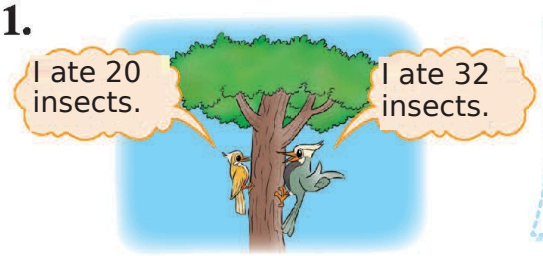
Practice

1.



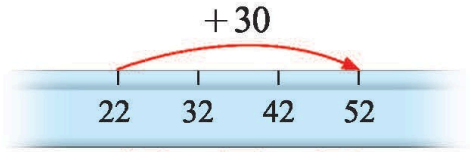
(1) How many insects were eaten in all?

(2) How many fewer insects did the small woodpecker eat than the big one?

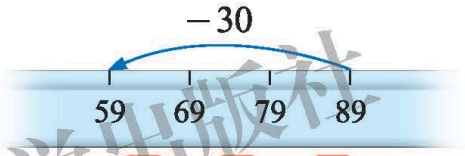
2. Show it with sticks, then calculate.

$68+30=$ $75-40=$

3. Look and fill in.



□ + □ = □

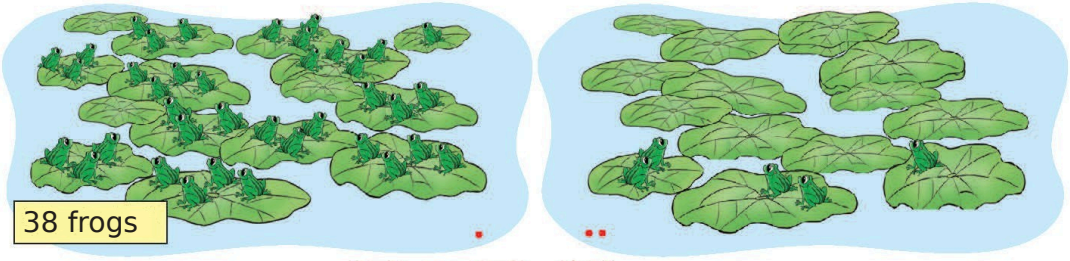


□ − □ = □

4.

$85-30=$	$36+50=$	$98-80=$	$74-50=$
$56+40=$	$45+20=$	$66-60=$	$30+48=$

5. How many frogs jumped away?



□ ○ □ = □

6. Each person gets 1 bottle of water. How many more bottles are needed?

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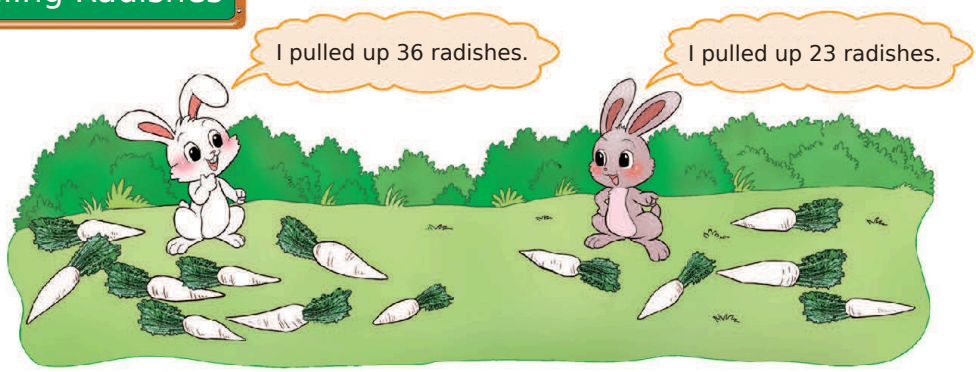
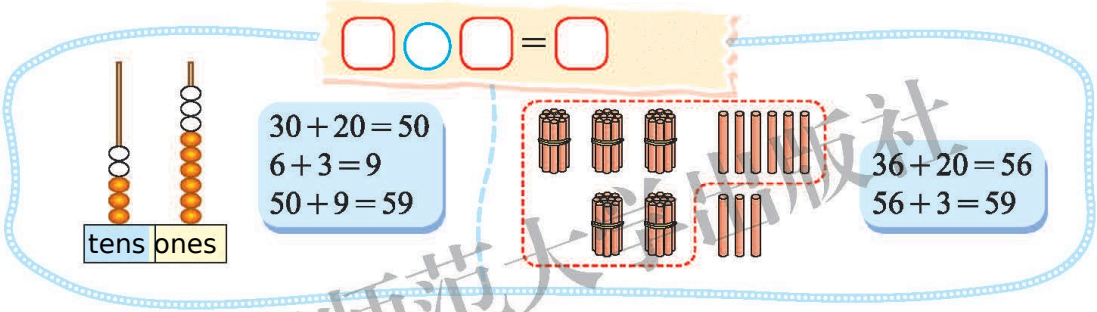
Pulling Radishes

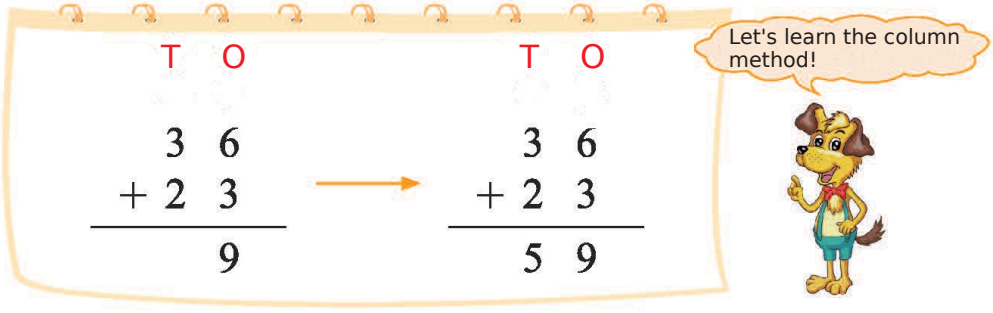


How many radishes were pulled up in all?



Answer: They pulled 59 radishes in all.

Learn and calculate.



Use the column method.

25 + 71 =

66 + 13 =

40 + 49 =

Practice

1. Move the beads on a counting frame and calculate.

$46+23=$ $56+42=$ $63+35=$

2. How many keys are there in all?

3. Use the column method.

$37+12=$ $5+63=$ $72+27=$

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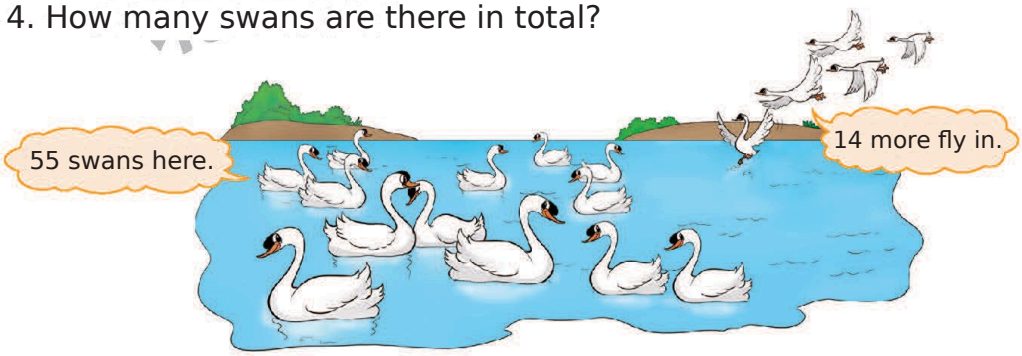
4. How many swans are there in total?

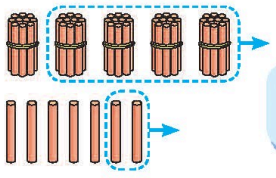


5. How many cabbages were harvested in all?

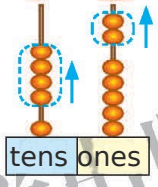
Harvesting Corn

How many fewer corn cobs did Xiaoxiao harvest than Taoqi?

□ ○ □ = □



57 − 40 = 17
17 − 2 = 15



50 − 40 = 10
7 − 2 = 5
10 + 5 = 15

Answer: Xiaoxiao picked ____ fewer corn cobs than Taoqi.

Can you use the column method? Think and fill in.

T	O
5	7
− 4	2
	□

→

T	O
5	7
− 4	2
□	□

Talk about it: where is the mistake?

6 5	3 4	7 6
− 3	− 1 1	+ 1 2
3 5	4 5	6 4

Line up ones with ones, tens with tens...

Calculate carefully...

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Practice

1. Move the beads on a counting frame and calculate.

$37-25=$ $42-11=$ $45-24=$

2. How many balloons haven't been blown up yet?

3. Use the column method.

$\begin{array}{r} 44 \\ +\ 32 \\ \hline \end{array}$	$\begin{array}{r} 54 \\ -\ 23 \\ \hline \end{array}$	$\begin{array}{r} 76 \\ +\ 23 \\ \hline \end{array}$	$\begin{array}{r} 68 \\ -\ 11 \\ \hline \end{array}$

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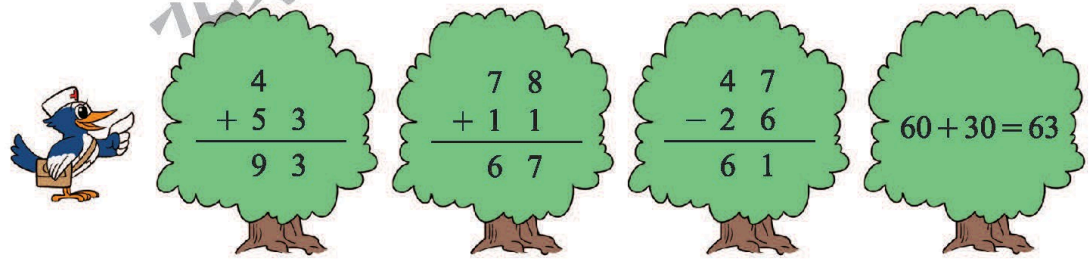
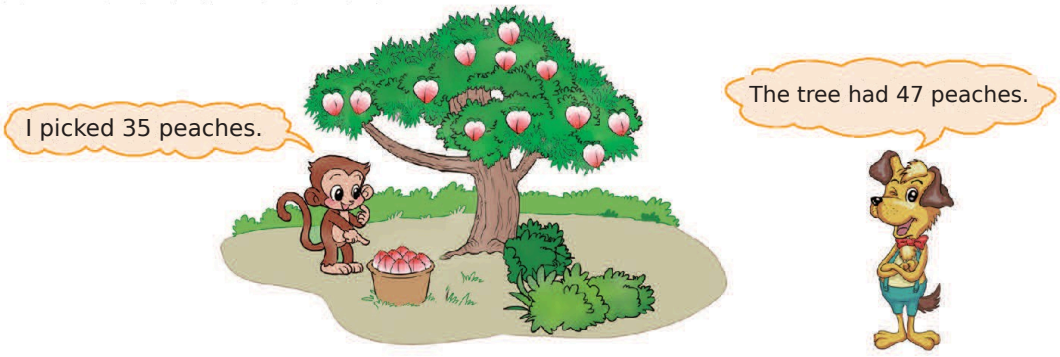
4. Forest Doctor



5. How many peaches are still on the tree?

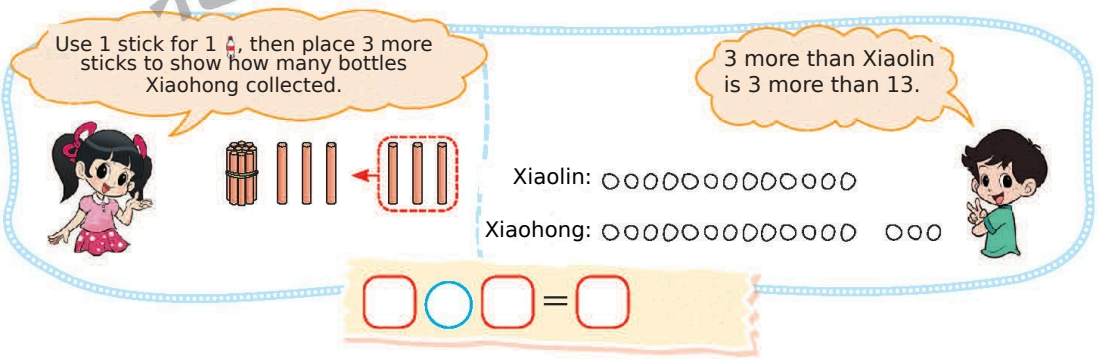


Recycling

Plastic buried in the ground won't "rot" for many years. Its damage to the environment is huge!

What maths problems can you ask?

How many 🍾 did Xiaohong collect?



Answer: Xiaohong collected ____ bottles.

How many 🍾 did Xiaoqing collect?

Answer: Xiaoqing collected ____ bottles.

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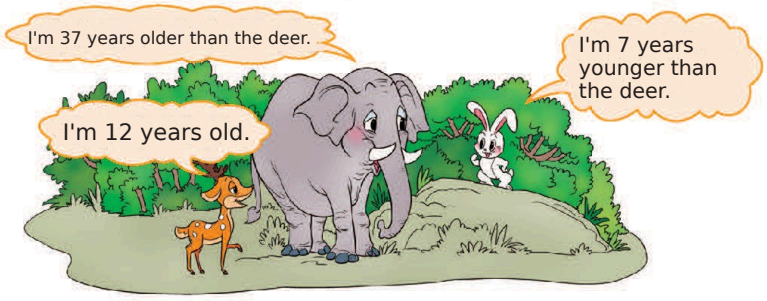
Practice

1. Look at the picture, ask two maths questions and solve them.



2. Draw and calculate.

(1) 5 more ○ than △.

△△△△△△△

_____

□○□=□

(2) 7 fewer △ than ○.

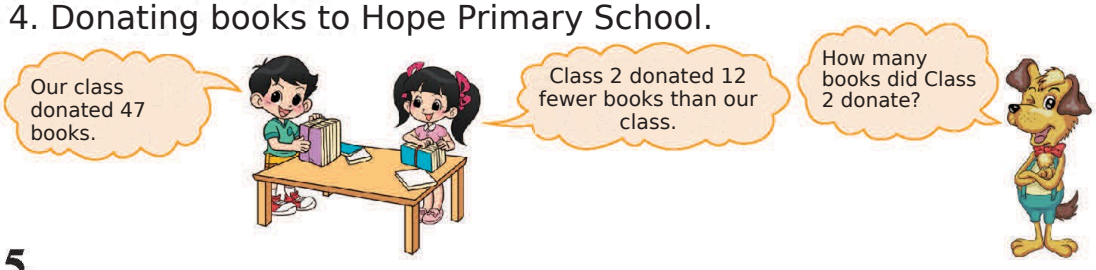
○○○○○○○○○○○

_____

□○□=□

3. How many stars did Xiaoxiao fold?

4. Donating books to Hope Primary School.



5.

$42+34=$	$34+5=$	$65-2=$	$78-60=$
$56-21=$	$40+23=$	$6+32=$	$59-4=$

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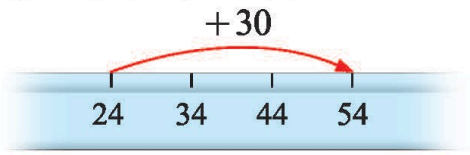
Exercise 3

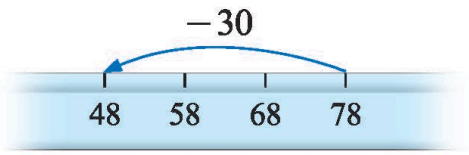
1. Move the beads on a counting frame and calculate.

$7+60=$ $76-14=$ $20+43=$ $57-3=$

2. Look and fill in.



□ + □ = □



□ − □ = □

3.

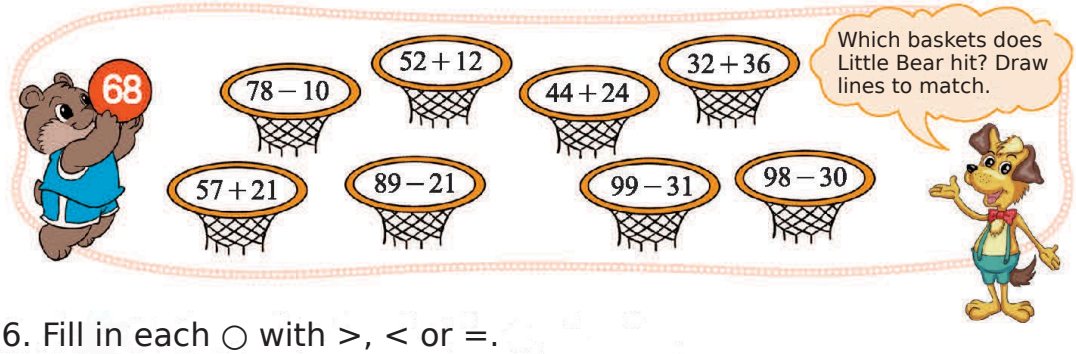
$30+50=$ $40+16=$ $67-30=$ $58-6=$

$80+16=$ $84-40=$ $14+22=$ $96-51=$

4.

$$\begin{array}{r} 78 \\ -\ 32 \\ \hline \end{array} \qquad \begin{array}{r} 65 \\ +\ 34 \\ \hline \end{array} \qquad \begin{array}{r} 56 \\ +\ 23 \\ \hline \end{array} \qquad \begin{array}{r} 99 \\ -\ 35 \\ \hline \end{array}$$

5. Little Bear Shoots Hoops



6. Fill in each ○ with >, < or =.

$87+10 \bigcirc 78+10$ $46-5 \bigcirc 46-4$ $71+6 \bigcirc 72+6$

$34+23 \bigcirc 41+16$ $86-40 \bigcirc 86-4$ $96-50 \bigcirc 6+50$

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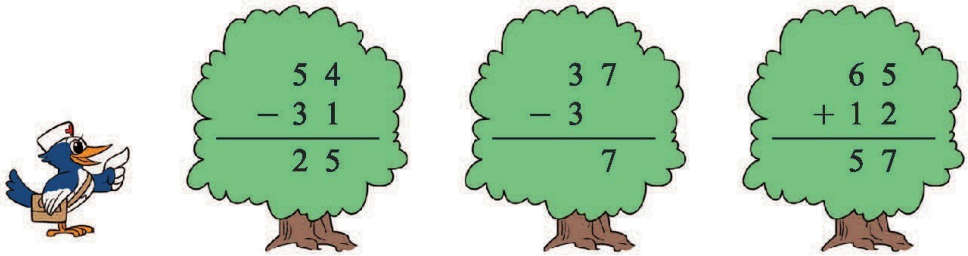
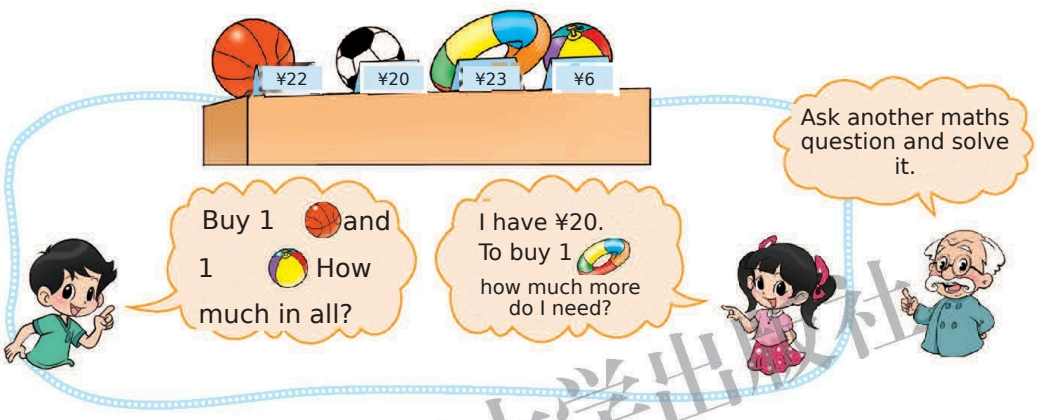
7. Forest Doctor

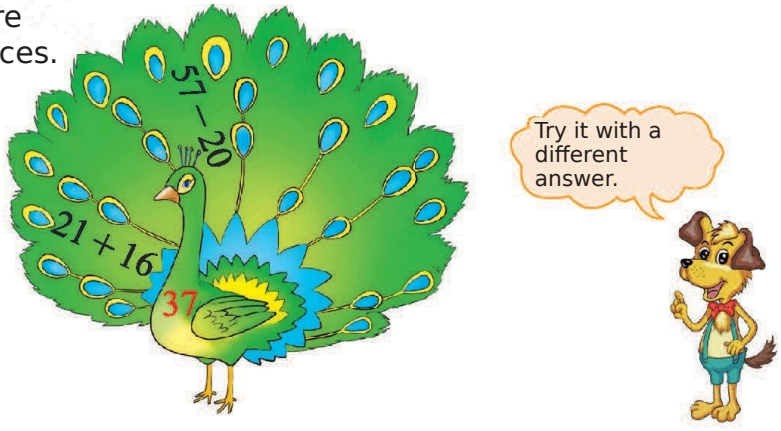


8.



9. Recycling used batteries

10. Write 3 more number sentences.



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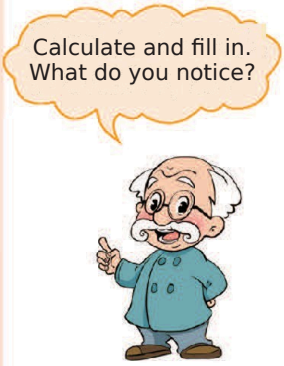
11. Mum has ¥100 and wants to buy an outfit. What could she buy? How much would she pay? Think of one way to do it.

12 The Naughty Puppy

$$\begin{array}{r} 2\square \\ -\ \square 2 \\ \hline 7 \end{array} \qquad \begin{array}{r} 9 \\ +\ 2\square \\ \hline \square 3 \end{array} \qquad \begin{array}{r} \square \\ -\ 4 \\ \hline 5\ 3 \end{array}$$

13 Magic Number Sentences

$11+11=\square$	$22-11=\square$
$12+21=\square$	$33-21=\square$
$13+31=\square$	$44-31=\square$
$14+\square=\square$	$55-\square=\square$
$\square+\square=\square$	$\square-\square=\square$
$\square+\square=\square$	$\square-\square=\square$
$\square+\square=\square$	$\square-\square=\square$
$\square+\square=\square$	$\square-\square=\square$



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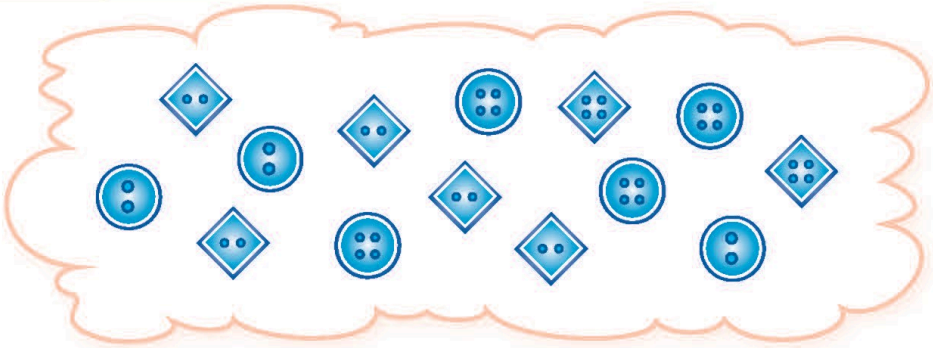
Maths Is Fun

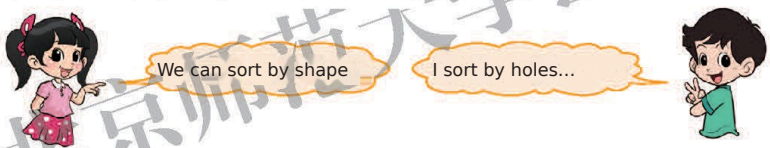
Sorting Buttons



Task: Sort the buttons in different ways.

Discuss

How will you sort them? Tell a partner.



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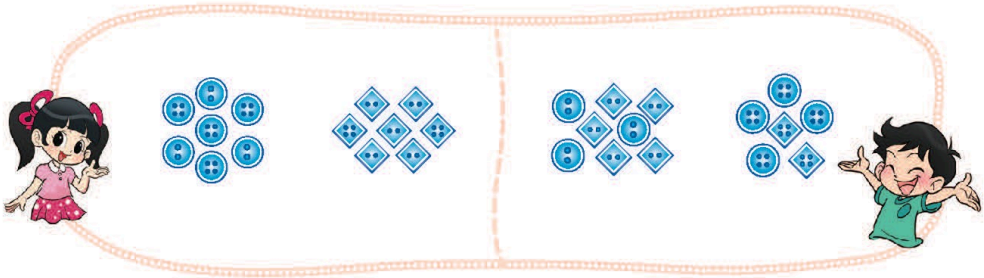
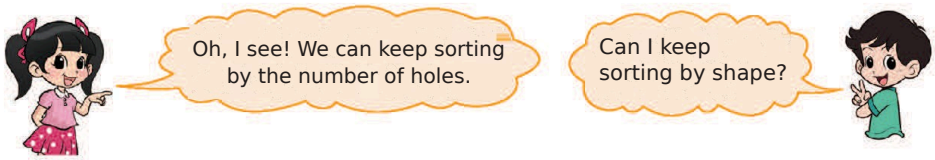
Do It

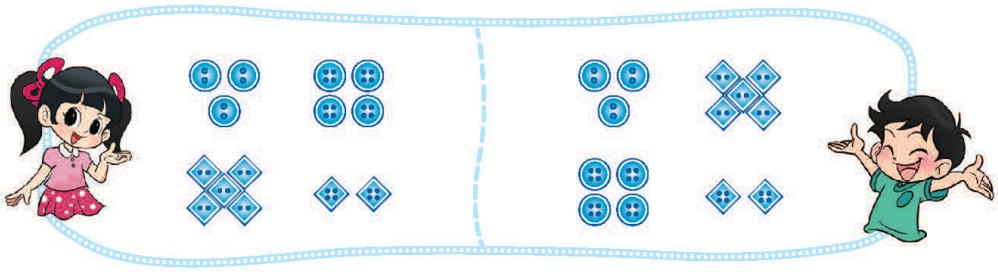
1. Sort the buttons above.



2. Talk with your partner. What new ideas do you have?

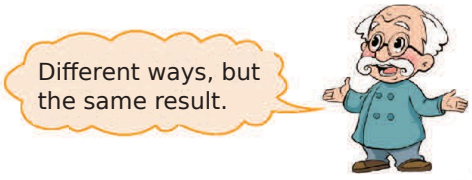


3. Here is how Taoqi and Xiaoxiao sorted. How did you sort?



Think

1. Compare how Taoqi and Xiaoxiao sorted. What do you notice?



2. Sort the buttons below in different ways and keep sorting. Are the results the same? Try it.

Self-check

How did I do in this activity?

I can find a suitable sorting method.			
I can take an active part in sorting.			
I can think hard during the activity.			

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Number Game

Work together to pass the levels.

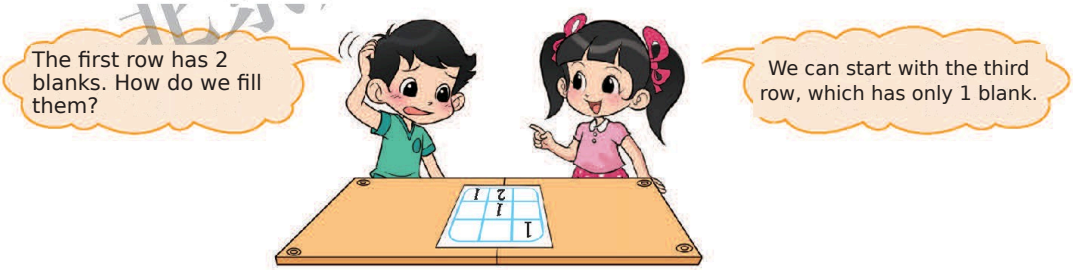
Rules of the game:

- Write one of 1, 2, 3 in each blank.
- Numbers must not repeat in any row or column.

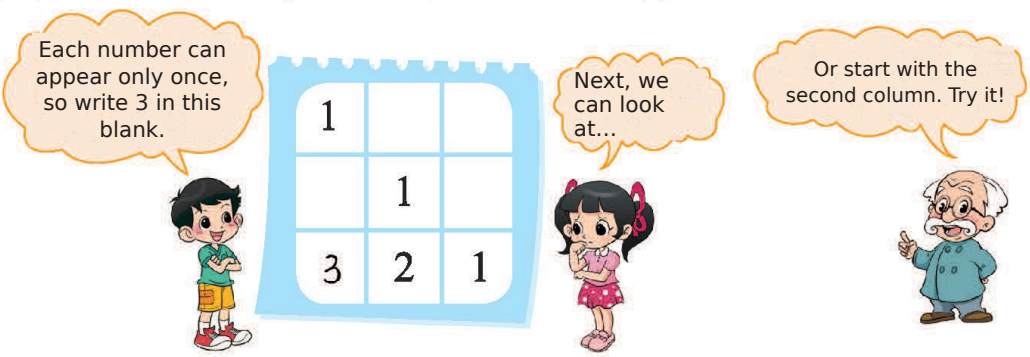
1		
	1	
	2	1

(1) Can you understand the rules? Explain them.

(2) Try it. Is it difficult?



(3) Keep going and explain your thinking.



Next Level

Same rules as above, but each blank can be one of 1, 2, 3, 4 or 5.

5	1			3
1	3			4
4	2		1	5
2		4	3	1
3	4	1		2

5	1			3
1	3			4
4	2	3	1	5
2	5	4	3	1
3	4	1	5	2

What should we do next?

The third cell in the first row can only be 2 or 4, but the third column already has 4, so it can only be 2...

Number maze: go from 51 to 100 in order. At each ★, fill in a number to get through.

End

62	85	86	90	91	94	92	95	100
63	★	87	88	89	90	★	94	99
60	83	82	81	80	81	92	93	98
59	60	61	64	79	94	93	96	97
58	69	62	79	78	77	94	★	94
57	★	63	64	65	76	77	96	95
57	55	54	55	66	★	74	93	96
51	52	53	68	67	69	73	98	97
52	54	54	69	70	71	72	99	98

Start

At ★ write 56 to pass the first level!

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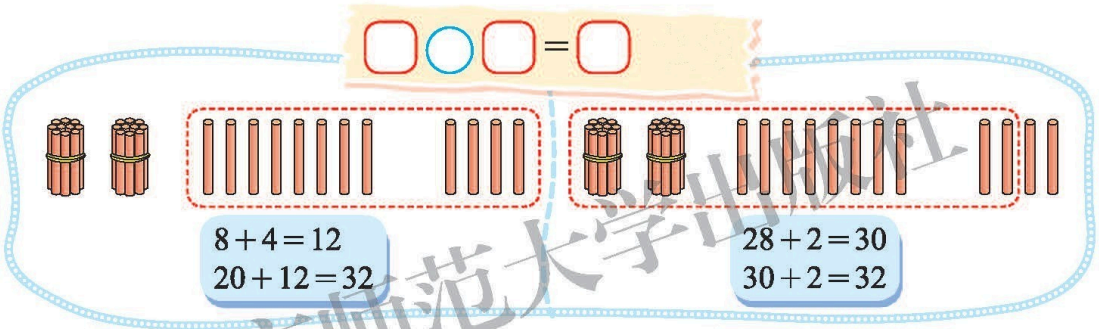
Unit 6 Addition and Subtraction (3)

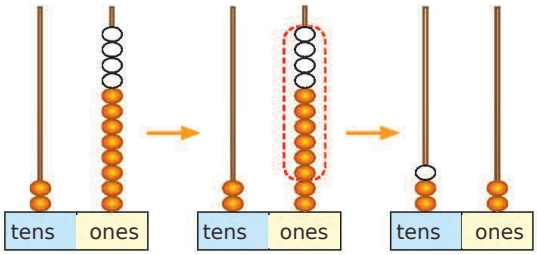
Library

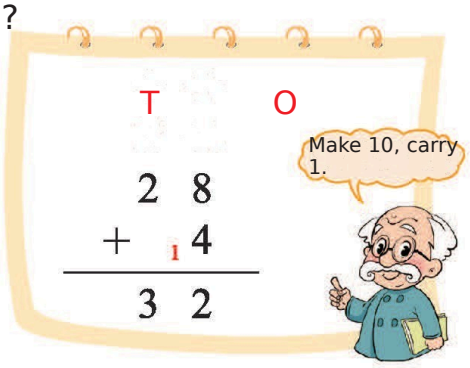
and — how many books in total? Use sticks, then calculate.



Answer: and have ____ books in total.

Can you use the column method?





and — how many books in total?

□○□=□

Answer: and have ____ books in total.

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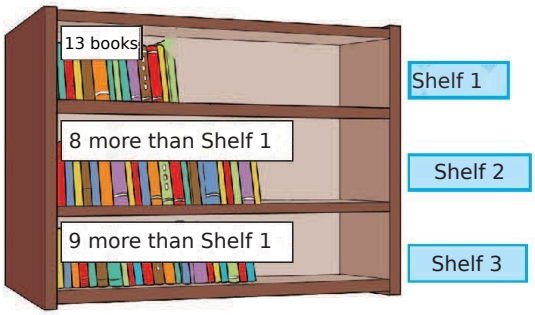
Try It



How many books are on the second shelf? Draw and fill in.

Shelf 1 ○○○○○○○○○○○○○○
Shelf 2

□○□=□

Answer: The second shelf has ____ books.

How many books are on the third shelf? Draw and calculate.

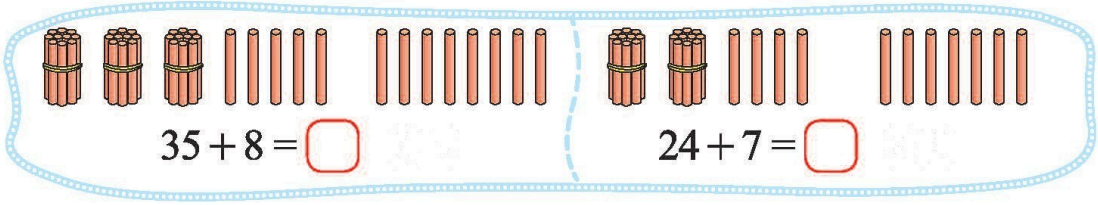
□○□=□

12 13 14 15 16 17 18 19 20 21 22 23

Answer: The third shelf has ____ books.

Practice

1. Circle and calculate.



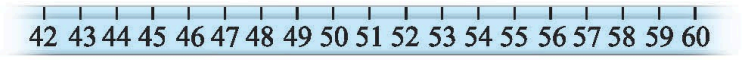
2. Use the column method.

58+7=	5+32=	38+6=	8+27=

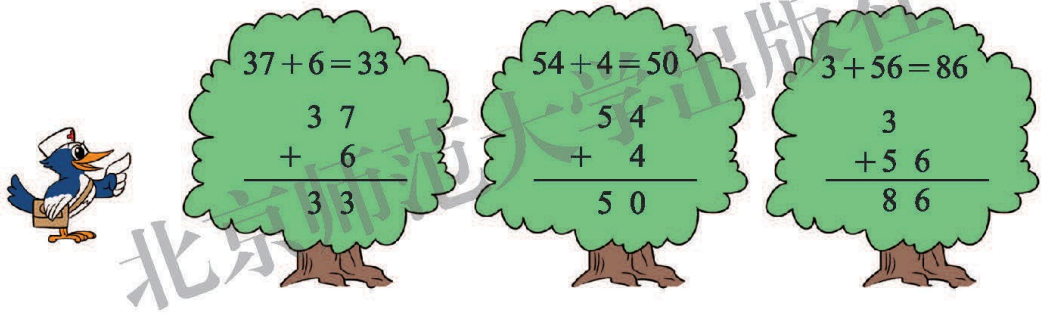
3. How many did Brother Rabbit pick?

4. Draw and calculate.

$47+9=\square$



5. Forest Doctor



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6. Fill in ◯ with ">", "<" or "=".

$12+9 \bigcirc 11+9$ $\quad$ $32+7 \bigcirc 7+32$

$67-4 \bigcirc 67+4$ $\quad$ $24+7 \bigcirc 24+70$

7. This row has $\square$ people in all.

Picking Apples

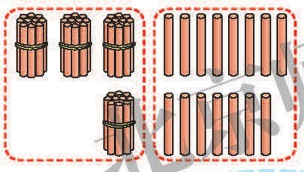
Estimate: can this box hold them?

Think of 38 as 40.
$17+40=57$,
so they fit.

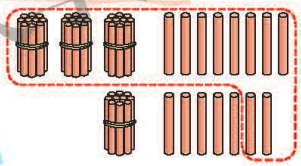
Think of 38 as 40 and 17 as 20. 40+20=60, so they fit.

How many apples in all? Calculate and discuss.

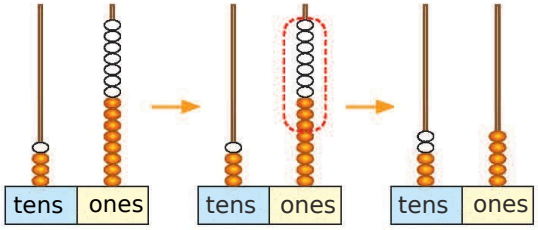
□ ○ □ = □ (apples)



$30+10=40$
$8+7=15$
$40+15=55$



$38+2=40$
$40+15=55$



T	O
3	8
+ 1	7
5	5

Answer: They picked ____ apples in total.

Calculate and discuss.

Line up place values; start from the ones.

$37+24=$

$46+54=$

When the ones reach ten, carry 1 to the tens; when the tens reach ten...

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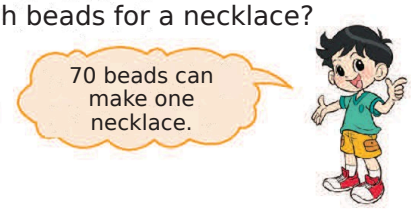
Practice

1. Estimate: which two bags together have enough beads for a necklace?

43 beads 19 beads 38 beads



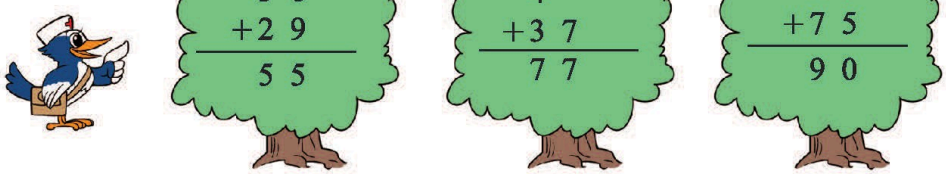
2. Use sticks or a counting frame, then calculate.

$27+24=$ $35+15=$ $69-11=$ $38+62=$

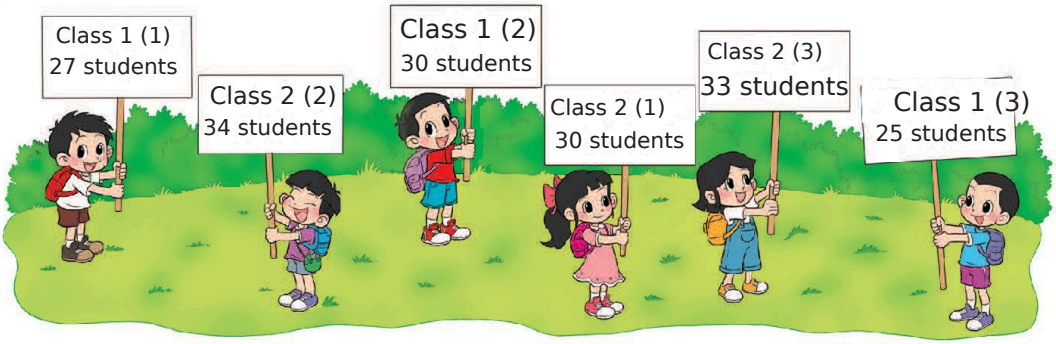
3. Use the column method.

$26+7=$	$18+27=$	$3+39=$
$26+27=$	$18+47=$	$23+39=$

4. Forest Doctor



5. Each bus holds at most 60 people. Which two classes best fit on one bus?



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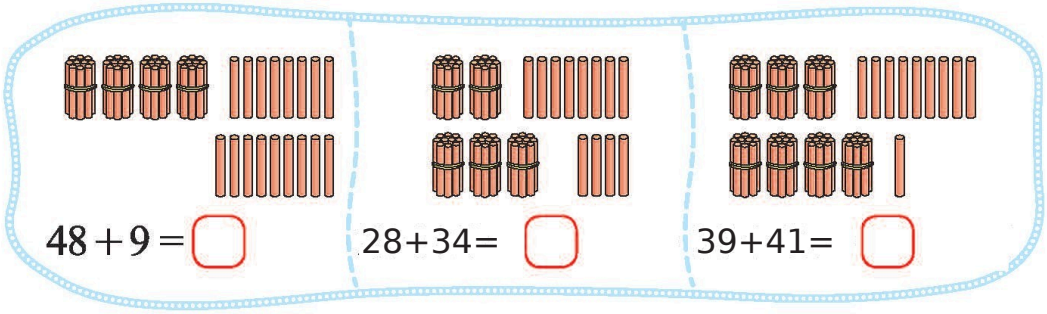
Exercise 4

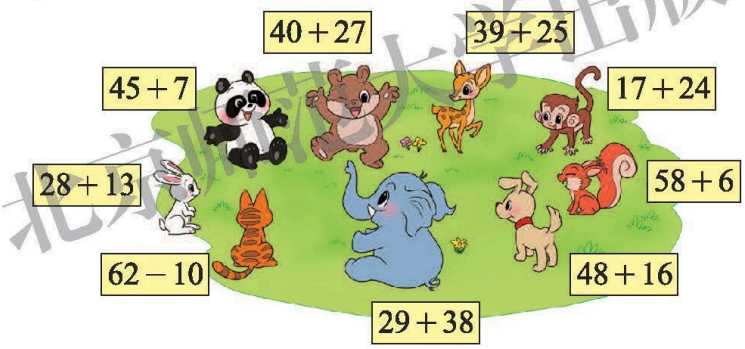
1. Circle and calculate.



2. Draw and calculate.

$43 + 9 =$ ☐

3. Find the pairs.



4. Use the column method.

$23 + 17 =$	$67 + 15 =$	$71 + 29 =$	$55 - 50 =$

5. Will all the teacups fit into the carton?

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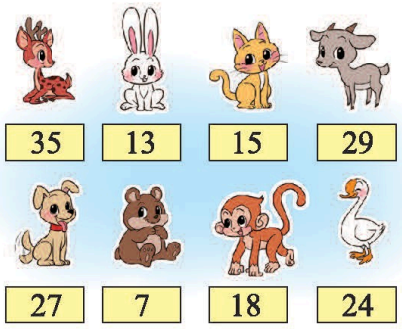
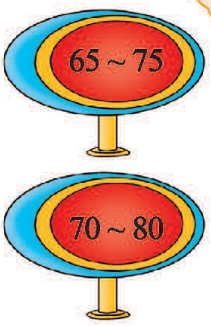
6. Taoqi ringed two animals and scored 42 points. Which two could they be?

Write two pairs.



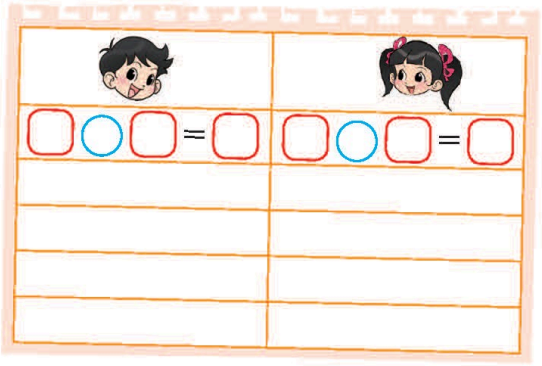
7.



20 + ____
35 + ____
16 + ____

50 + ____
38 + ____
44 + ____

8. Maths game



Two players take turns to pick cards.

(1) On each turn, take two cards and add the numbers. The sum must be less than 100.

(2) Fill in the table; each round, circle the larger of the two sums.

(3) Play 5 rounds without reusing cards. Whoever has more circles wins.

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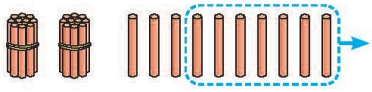
Reading Room

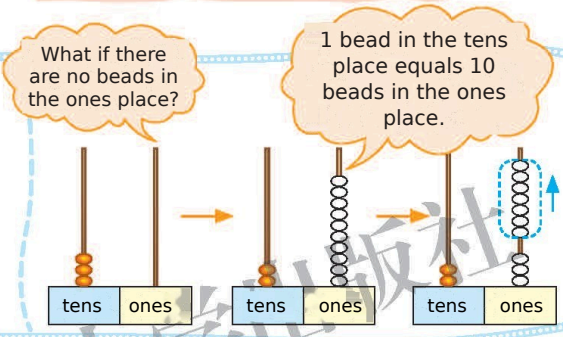
	Magazines	Stories	Comics
Had	30	25	46
Lent	7	8	9

How many magazines are left?

□ ○ □ = □

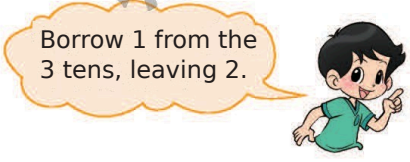


$10-7=3$
$20+3=23$

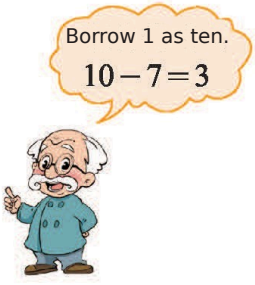


Answer: ____ magazines are left.

Can you use the column method?



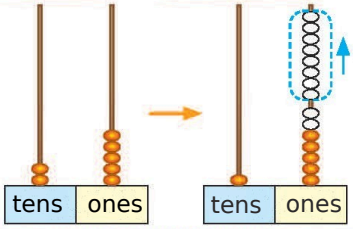
T	O
3	0
−	7
2	3



How many storybooks are left?

□ ○ □ = □

$15-8=7$
$7+10=17$



T	O
2	5
−	8
1	7

Answer: ____ storybooks are left.

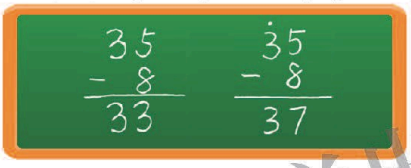
Try It

How many ears of corn did 🐻 pick?

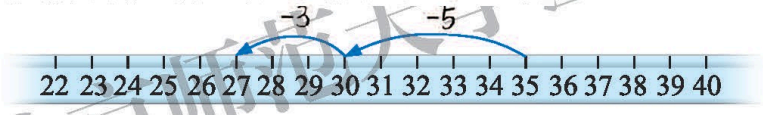
□ ○ □ = □

Answer: 🐻 picked ____ ears of corn.

Discuss: are these calculations correct? How can we fix them?



Taoqi calculated it like this. Look and talk about it.



Practice

1. Use sticks or a counting frame, then calculate.

42 − 6 =	50 − 3 =	54 − 8 =

2. Think and calculate.

10 − 7 =	13 − 5 =	100 − 1 =
30 − 7 =	23 − 5 =	100 − 2 =
50 − 7 =	33 − 5 =	100 − 3 =

3. Use the column method.

48 − 9 =	53 − 6 =	70 − 7 =	32 − 3 =

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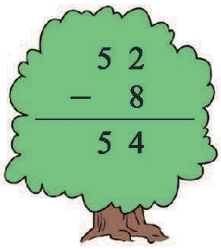
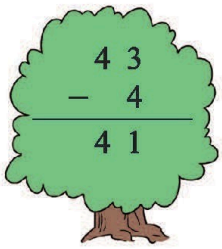
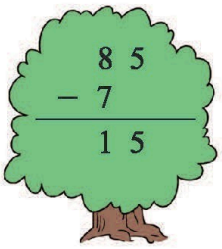
4. Forest Doctor



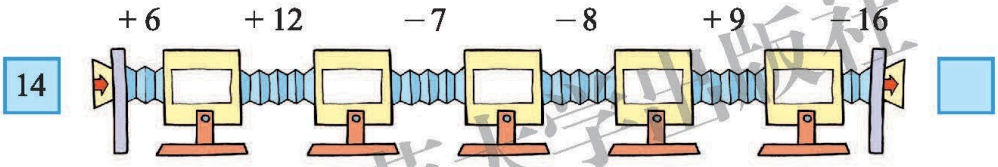




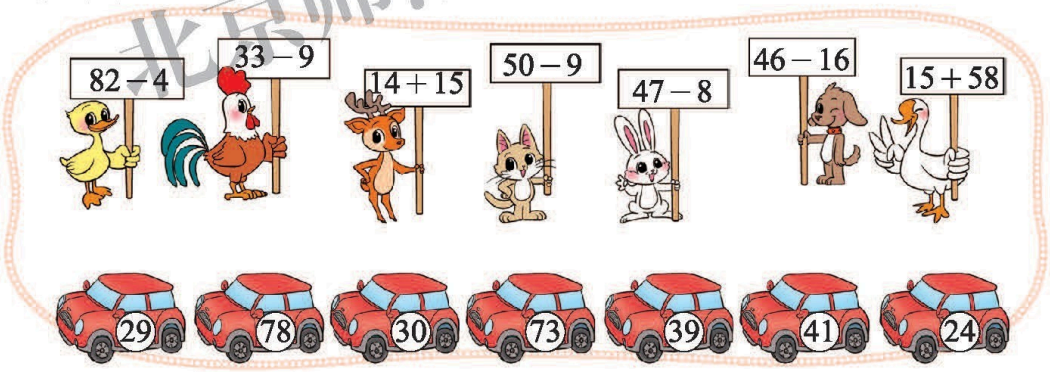
5.

$40-9=$	$81-4=$	$73-8=$	$56-7=$
$62-6=$	$9+45=$	$46-8=$	$36+5=$
$49+27=$	$91+9=$	$85-15=$	$32-8=$

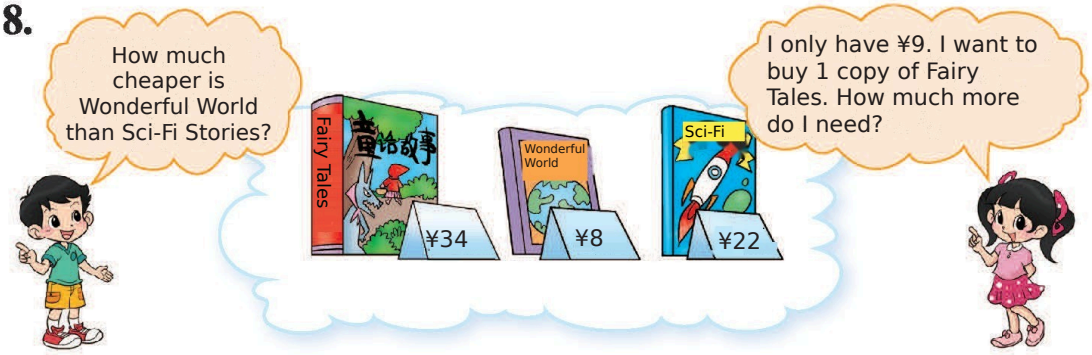
6.



7. Find each animal's car.



8.



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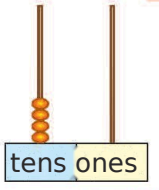
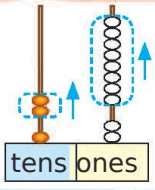
Skipping

Xiaohong	Xiaoliang	Xiaodong
40 skips	28 skips	32 skips

How many more skips did Xiaohong do than Xiaoliang?

□ ○ □ = □ (skips)





T O

$$\begin{array}{r} 4\ 0 \\ -\ 2\ 8 \\ \hline 1\ 2 \end{array}$$

Answer: Xiaohong did ___ more skips than Xiaoliang.

How many fewer skips did Xiaoliang do than Xiaodong?

□ ○ □ = □ (skips)

T O

32 − 20 = 12
12 − 8 = 4

$$\begin{array}{r} 3\ 2 \\ -\ 2\ 8 \\ \hline 4 \end{array}$$

Answer: Xiaoliang did ____ fewer skips than Xiaodong.

Calculate and discuss.

Line up the same places and start from the ones.

21 − 2 =

53 − 6 =

70 − 34 =

Not enough ones? Borrow 1 ten.

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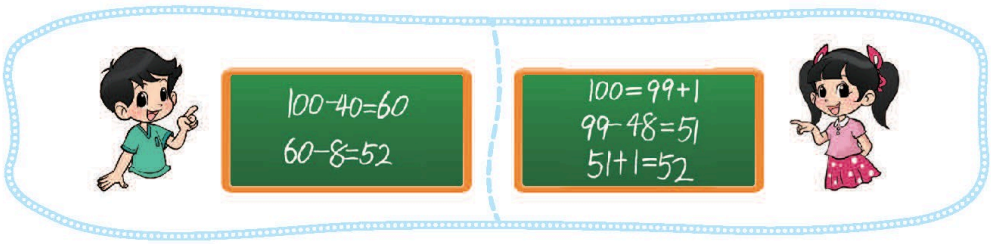
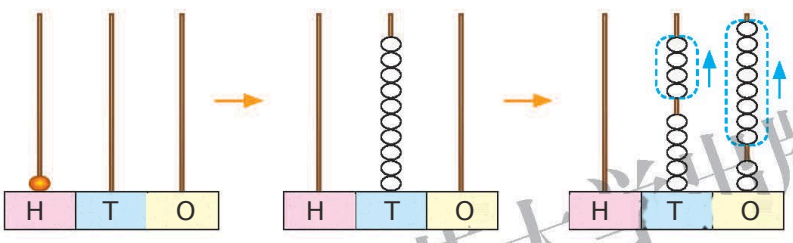
Try It

100 − 48=? Can you understand their methods?



Move the beads, then try using the column method.



H	T	O
1	0	0
−	4	8
	5	2

Practice

1. Use sticks or a counting frame, then use the column method.

85 − 58 =

44 − 27 =

80 − 55 =

64 − 19 =

2. How much does 1 badminton racket cost?

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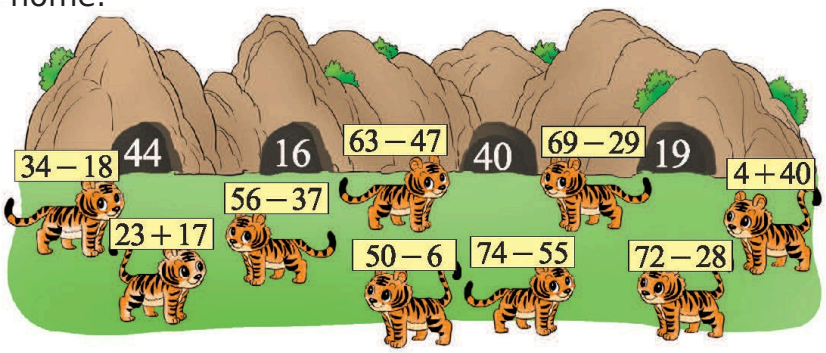
3. Calculate, and explain your method.

$100-53=$ $100-18=$

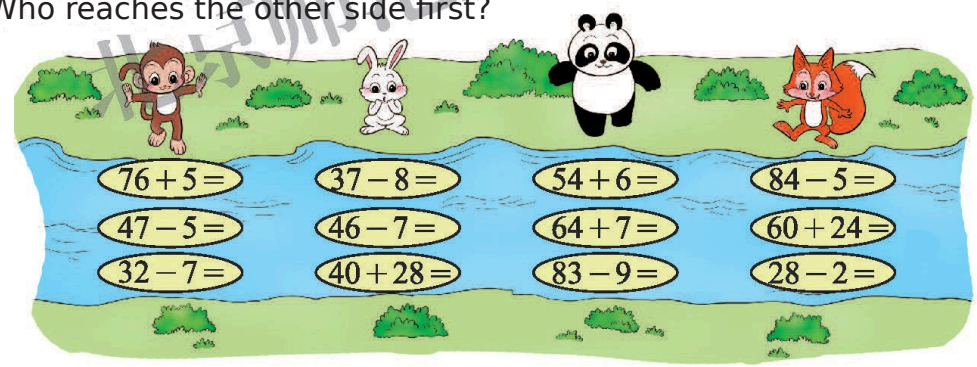
4. Going home.



5.

$53+21=$	$65-15=$	$34-26=$
$72-9=$	$32-17=$	$80-16=$
$5+39=$	$63-8=$	$100-39=$

6. Who reaches the other side first?



7. Athletes, line up!

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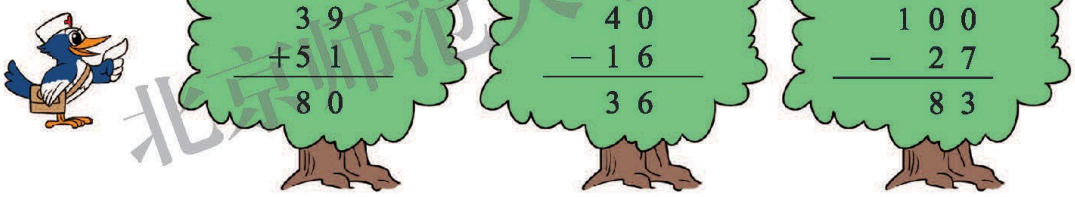
Exercise 5

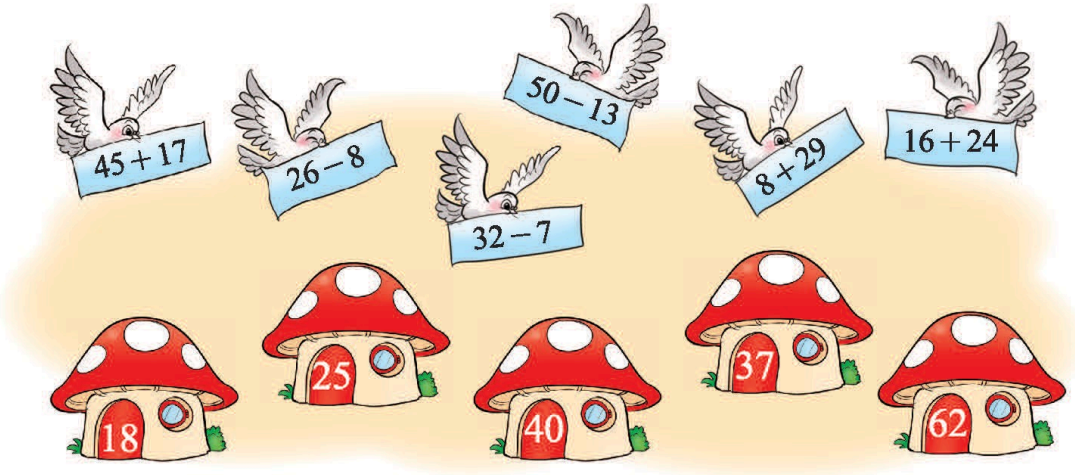
1. Use sticks or a counting frame, then use the column method.

$90-47=$	$57-19=$	$100-28=$

2. Forest Doctor



3. Going home.



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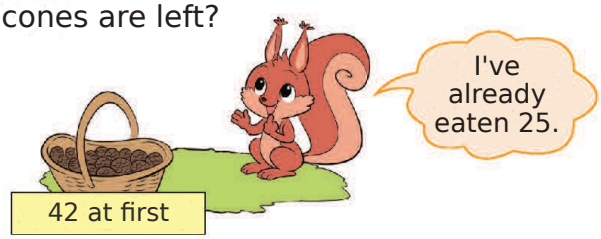
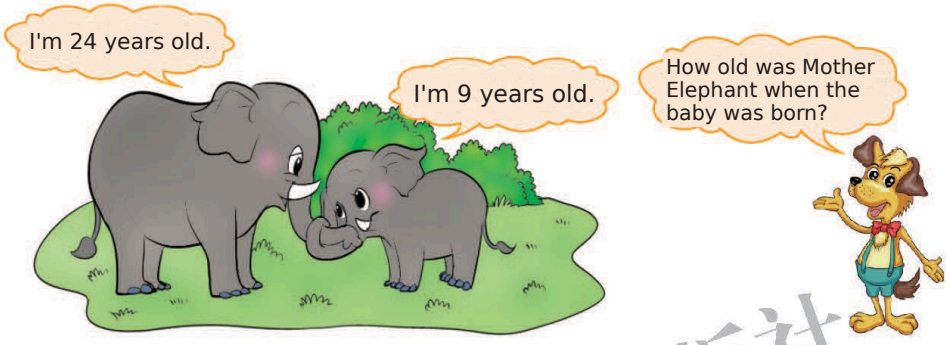
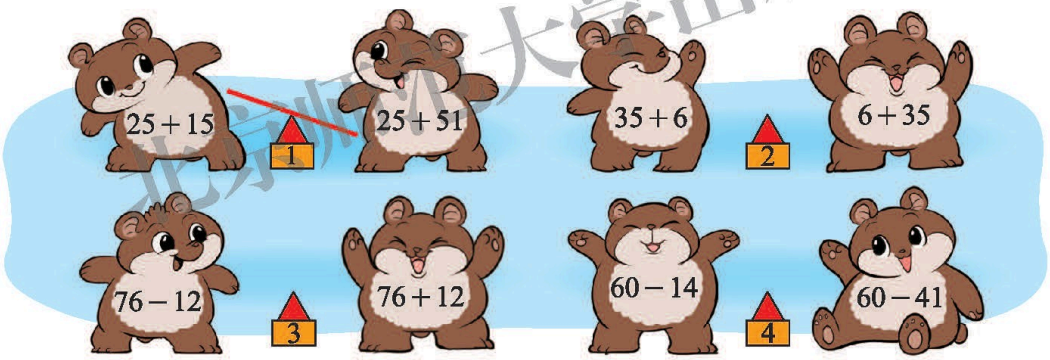
4. How many pine cones are left?



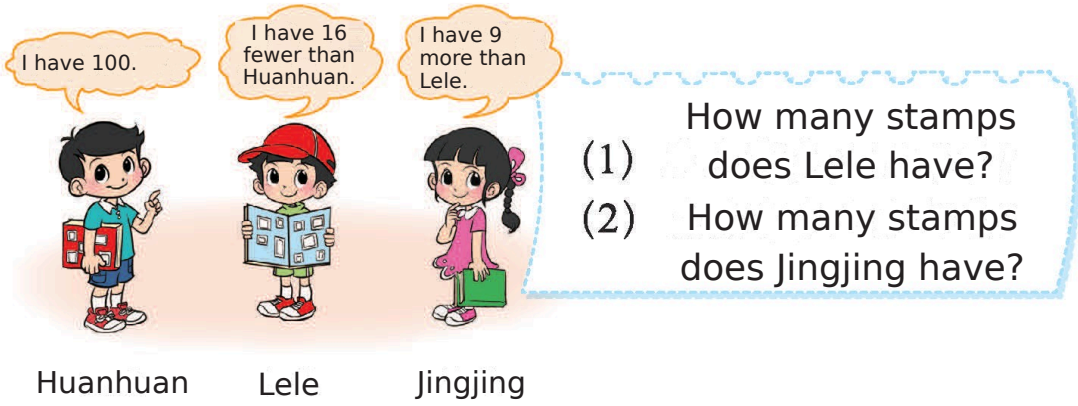
5.



6. Draw the seesaws.



7. Collecting stamps.



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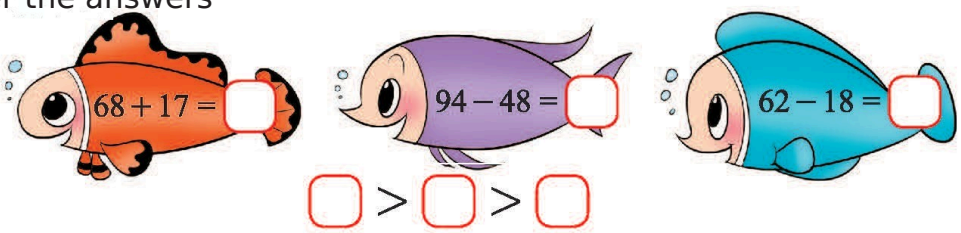
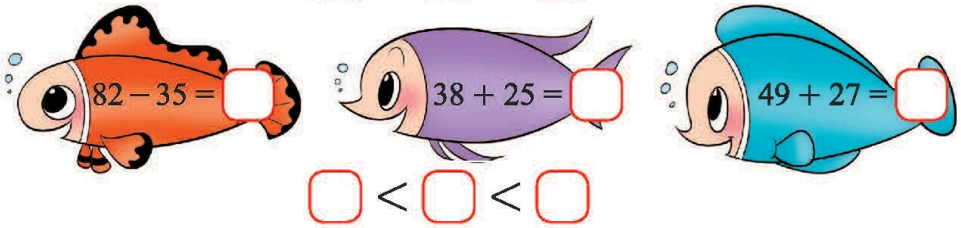
8. Order the answers

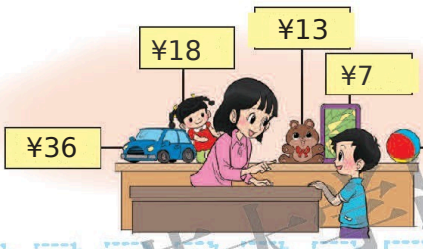
(1) 

□ > □ > □

(2) 

□ < □ < □

9.



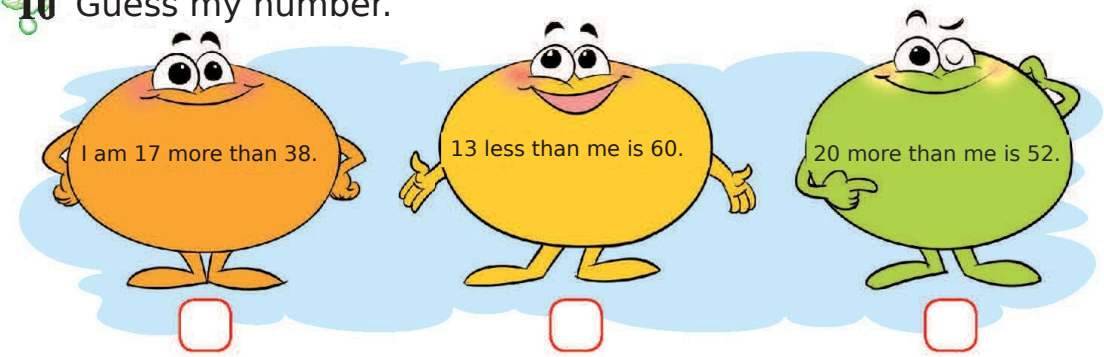
(1) Xiaoming has ¥50. Which three toys can he buy? Circle them.

(2) How much more does a toy car cost than a doll?

(3) Xiaoliang bought a toy car and got ¥64 change. How much did he give the salesperson?

(4) Ask another maths question and solve it.

10 Guess my number.



Final Review

Review and Share

Number and Algebra

Think alone

1. What numbers do you know? Can you count them in order?

2. How many different ways can you show the number 32?

3. How do you compare two numbers? Give an example.

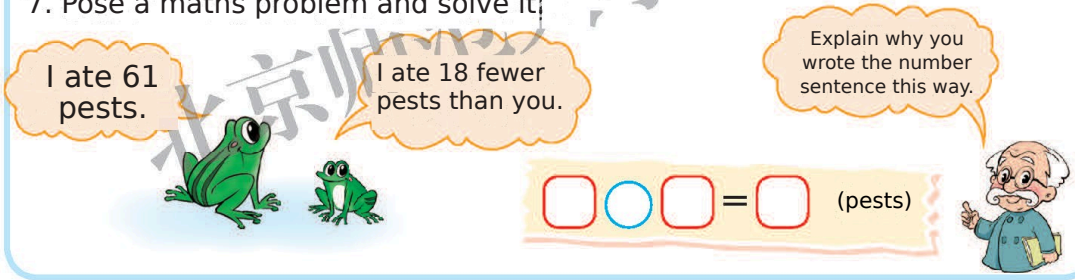
4. Use the column method, then explain how with sticks or a counting frame.

$45+27=$ $65-18=$ $73+27=$ $100-19=$

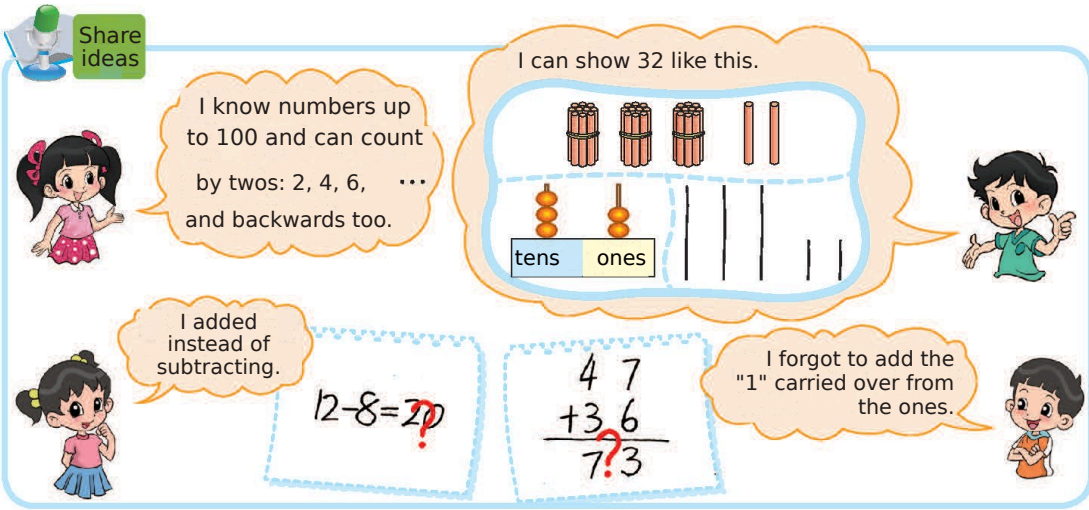
5. List problems you often get wrong and say what to watch out for.

6. What real-life problems can "50 − 18" solve? Give two examples.

7. Pose a maths problem and solve it.



Share ideas



Shapes and Geometry

Think alone

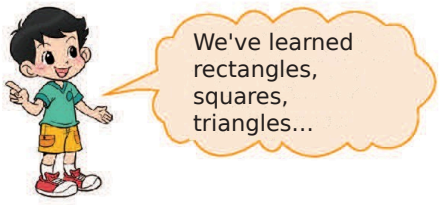
1. Which shapes have you learned this term? Describe them in your own words and try to draw them.

2. Which shapes make up each pattern below? Can you design an interesting pattern?

Skating Football Dancing

3. What does each child see? Draw lines to match.

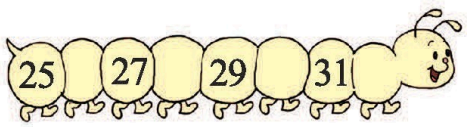
Share ideas

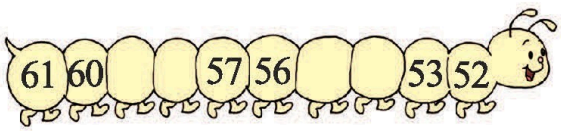


Exercise

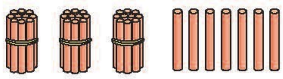
Number and Algebra

1. Fill in.



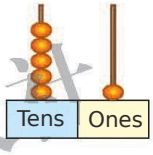


2. Look and fill in.



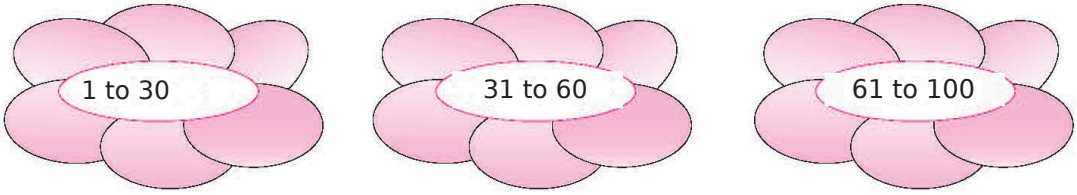
37 has ( ) tens and ( ) ones.

24 has ( ) tens and ( ) ones.



51 has ( ) tens and ( ) ones.

3. Match.



4.

$40+30=$	$45+7=$	$69-50=$	$97-6=$
$30+65=$	$9+36=$	$53-8=$	$84+7=$

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5. Use the column method.

$58+37=$	$94-35=$	$85+15=$	$98-89=$

6. Fill in ◯ with >, < or =.

$34+29$ ◯ $29+34$　　$53-36$ ◯ $54-36$

$49-37$ ◯ $49-36$　　$57+38$ ◯ $57+34$

7.

A group of ducks is on the river. There are 27 ducklings, and there are 15 fewer big ducks than ducklings. How many big ducks are there?

8. Calculate and discuss.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

$15+16=$
$25+26=$

$15+25=$
$16+26=$

$15+26=$
$16+25=$

What do you notice? Try circling four numbers in the table like the example.

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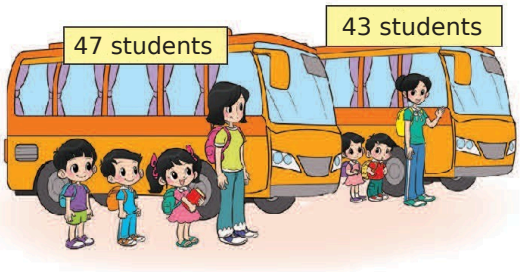
9. Two groups go to summer camp. Each group has 3 teachers, and each bus seats 50 people.



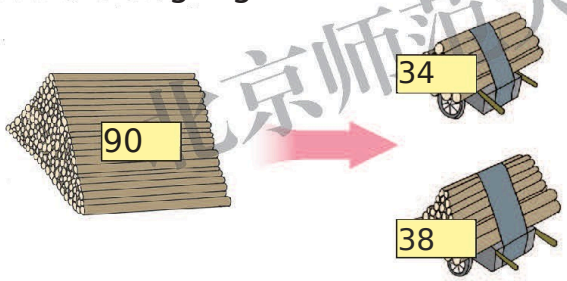
Need ☐ buses.

Left: ☐ empty seats.

10.

Ask two maths questions and try to answer them.

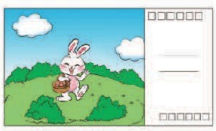
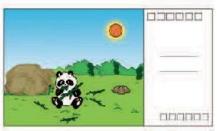
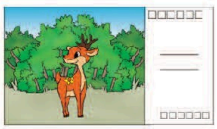
11. Moving logs.



(1) How many logs in all?

(2) How many are left?

12.

			Total
23 cards	38 cards		92 cards

(1) and – how many in total?

(2) – how many are there?

(3) Ask another maths question and try to answer it.

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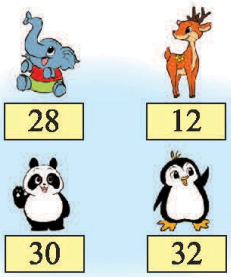
13. Ring toss game.



(1) Which two did each of them ring?

(2) Which two could Miaoxiang have ringed?

14.

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Rope-skipping record

Name	Liu Ping	Zhang Xiao	Li Yue	Wang Li
Score	76 skips	95 skips	83 skips	100 skips
Rank				

(1) Work out their ranks and write them in the table.

(2) Most: ___ skips. Fewest: ___ skips. Difference: ___.

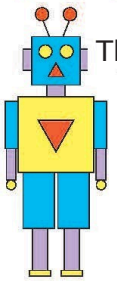
(3) Second place did ___ more skips than third place.

(4) Fourth place needs ___ more skips to match third place.

(5) Ask another maths question and try to answer it.

Shapes and Geometry

1. Find and count.



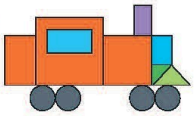
There are ____ ▭,

____ □,

____ △,

and ____ ○.

There are ____ ▭,



____ □,

____ △,

and ____ ○.

2. Who sees each picture below? Match.

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3. Fold the paper rectangle in Figure 4 of the appendix.

(1) Divide it into two triangles.

(2) Divide it into two rectangles.

(3) Divide it into one square and one rectangle.

4. Can you make a picture like the one below with the flat shapes you know?

What Did You Learn This Term?

1. _____

2. _____

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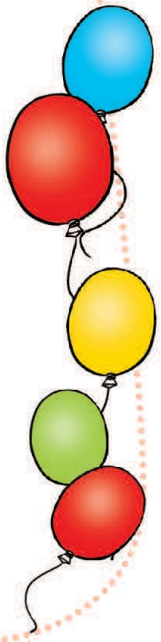
3. _____



Question Bank

What maths problems have you noticed in everyday life? Write them down. Can you solve them?

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Appendix

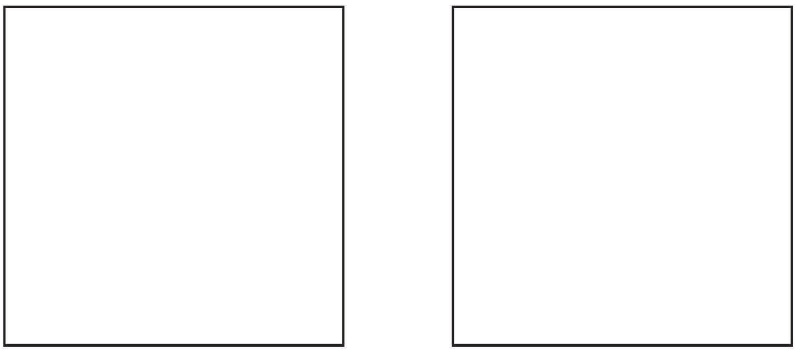

Figure 1

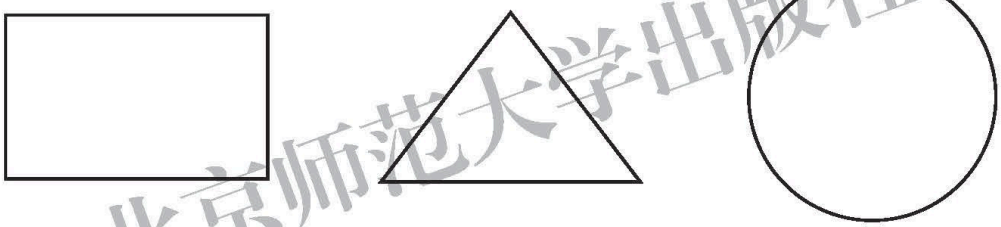

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Figure 2

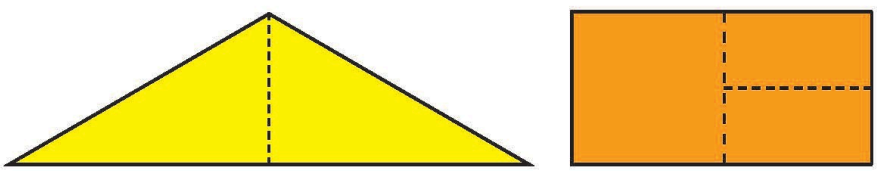

Figure 3

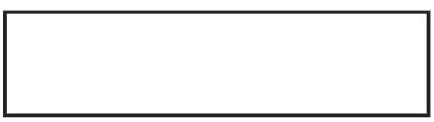

Figure 4

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Afterword

The Compulsory Education Textbooks (Beijing Normal University Edition) were written by many heads and core members of the groups that drew up the national curriculum standards for basic education, together with subject experts, education experts, psychologists and master teachers. They rest on solid research and advanced educational ideas, and offer high-quality writing and professional service. The textbooks strive to reflect the spirit of the national curriculum standards for basic education, make use of many kinds of information resources, show the latest progress in each subject where appropriate, and stress the use of knowledge, skills and ways of thinking in real life. They stay close to students' lives, pay attention to how students learn, meet their different learning needs and promote the all-round development of every student.

The Compulsory Education Textbooks (Beijing Normal University Edition) · Mathematics (Grades 1–6), also known as the New Century Primary School Mathematics Textbooks, fully embody the basic ideas of the new century's reform of the mathematics curriculum, with the healthy growth of children as their fundamental aim. This is the 4th edition of the New Century Primary School Mathematics Textbooks. Building on the strengths of the first three editions, it has been fully revised in accordance with the Mathematics Curriculum Standards for Compulsory Education (2011 Edition).

The new edition has the following distinctive features. Its basic form of presentation is "a situation plus a series of questions", which seeks to make the unfolding of the content, the students' learning, the teachers' teaching and the reaching of the curriculum goals one whole, good for learning and for teaching, so that students keep going through the process of thinking a problem through "from beginning to end". It seeks to give students realistic, interesting and challenging learning materials, to give them time and space to explore and to share their ideas, to show how mathematical knowledge is formed and used, and to meet the needs of different students.

The aim is for students to: appreciate the close links between mathematics, nature and human society; gain the basic knowledge, basic skills, basic ideas and basic experience of activities that they need and that suit their age; develop the ability to find and pose problems and to analyse and solve them; understand the value of mathematics, become more interested in learning it and more confident of learning it well; and gain a first sense of innovation and a scientific attitude.

The chief editors of the New Century Primary School Mathematics Textbooks are Liu Jian, Kong Qiping and Zhang Dan. The other members of the writing group are Hu Guangti, Liu Keqin, Wang Yong, Tao Wenzhong, Wang Mingming, Zhu Dejiang, Chen Xiaomei, Zhang Hong, Qian Shouwang, He Fengbo, Hua Yinglong, Huang Lihua, Zhao Yanhui, Li Tie'an, Wei Huinü, Zhu Yuhong, Ren Jingye and Hou Huiying.

The chief editors of this volume are Zhang Hong and Liu Keqin. The main people who took part in revising and finalising it were Chen Xiaomei, Hou Huiying, Zhu Dejiang, Huang Lihua, Wang Mingming, Ren Jingye, Fang Liping and Xiao Xiaoyu; the final text was compiled and approved by Liu Jian and Zhang Dan. Many local teaching researchers and front-line teachers gave valuable advice on this new round of revision.

Special thanks go to the children Liu Xiaohan, Liu Yiqian, Zou Chenye, Li Ruitong, Ding Muyun, Yin Jiaxin, Ma Changle, Luo Dezhi, Zhang Jingyi, Zhang Xiaoda, Han Leran, Mao Meiyi, Chen Zhao'ai, Zhao Yunqi, Ai Sente, Li Zihan, Li Jiaxuan, Wang Duo, Jiang Han and Zhang Zhimeng, and to Zhu Lei, who were invited to contribute their work.

We hope that teachers and students will send us their valuable comments as they use the book, so that we can revise and improve it further.

You are welcome to contact us by phone or letter: Primary School Mathematics Editorial Office, Beijing Normal University Press (100088), (010) 58802836, xxsx@bnupg.com; Mathematics Studio, Basic Education Curriculum Research Centre, Beijing Normal University (100082), (010) 58435911, xsjmath@126.com.

Beijing Normal University Press